

Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions

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(Dated:)

We consider the Quantum Approximate Optimization Algorithm at fixed low depth, where the dominant cost is the number of circuit evaluations needed to find good variational angles. A graph isomorphism network is trained to output a Gaussian distribution over angles for each input graph. The mean serves as an initial point for a derivative-free optimizer, the covariance defines a trust region that constrains the search, and the trace of the covariance sets an instance-dependent evaluation budget.

We prove bounds on the objective degradation inside the trust region, on gradient-variance lower bounds that prevent flat directions from stalling the search, on preservation of the expected objective ordering under depolarizing noise, and on finite-sample coverage of the predicted region. Together these guarantees ensure that confining the optimizer to the learned ellipsoid does not forfeit solution quality.

We test the method on MaxCut at depth $p=2$ for Erdős–Rényi, 3-regular, Barabási–Albert, and Watts–Strogatz graphs with 8 to 16 vertices. The mean evaluation count drops from 343 (random restarts) and 85 (best learned-point baseline) to 45 ± 7 , with approximation ratios within 0.03 of concentration-based heuristics. Total CNOT-gate executions are reduced by 87%. The predicted uncertainty is calibrated (ECE=0.052, Spearman $\rho=0.770$), the trust regions transfer across graph sizes without retraining, and the speedup persists from $7.2\times$ to $7.7\times$ under shot noise ranging from 512 to 8192 shots. At a higher budget operating point the ratio gap to the best absolute-quality baseline narrows to 0.007. The savings hold across graph families, sizes, seeds, and shot-noise levels.

I. INTRODUCTION

The Quantum Approximate Optimization Algorithm (QAOA) [1] applies p alternating layers of a cost unitary and a mixer unitary to the uniform superposition. The $2p$ variational angles are then chosen to maximize the expected value of the cost function. At low depth, typical derivative-free optimizers require hundreds of objective evaluations per instance [2, 3], and on quantum hardware each evaluation demands repeated state preparation and measurement. The number of circuit evaluations is therefore a dominant cost [4, 5].

We describe a method that uses a learned, graph-conditioned Gaussian to reduce this cost. A Graph Isomorphism Network (GIN) [6] with Laplacian spectral encodings [7, 8] takes a graph $\mathcal{G}$ and outputs a distribution $\mathcal{N}(\boldsymbol{\mu}(\mathcal{G}), \boldsymbol{\Sigma}(\mathcal{G}))$ over QAOA angles. The mean $\boldsymbol{\mu}$ initializes local optimization. The covariance $\boldsymbol{\Sigma}$ defines a Mahalanobis ellipsoid—the trust region—to which the optimizer is confined. The trace of $\boldsymbol{\Sigma}$ sets the number of evaluations allocated to the instance. A single learned object therefore specifies the initialization, the search boundary, and the computational budget. No external hyperparameter couples these three decisions; they all derive from the same predicted Gaussian.

The Gaussian prediction differs from prior warm-start and transfer approaches in that the optimizer is confined to the predicted region throughout, not merely initialized

by it, and the budget varies from instance to instance according to the model’s own confidence. The confinement prevents the optimizer from wandering into regions of the landscape that the model has no reason to trust; the instance-dependent budget avoids spending circuit evaluations on graphs the model already understands well.

Section V proves that the trust region preserves objective quality, that the gradient does not vanish inside the ellipsoid, that the expected ordering survives depolarizing noise, and that conformal calibration gives finite-sample coverage.

We evaluate the method on MaxCut at depth $p=2$. The test graphs are Erdős–Rényi, 3-regular, Barabási–Albert, and Watts–Strogatz families with $n=8$ –16 vertices. Training uses 240 graphs at $n=14$ only. The learned trust region transfers to families and sizes not seen during training, a consequence of the spectral positional encodings and the contrastive loss that structures the graph embedding by landscape similarity rather than by combinatorial label. On $n=14$ graphs the method reduces the mean evaluation count from 343 (random restarts) and 85 (best learned point predictor) to 45 ± 7 , while keeping sampled approximation ratios within 3 percentage points of concentration-based heuristics. The factor-of-two reduction relative to the point predictor is not explained by a better initialization; it arises from the trust-region constraint and the adaptive budget, which the point predictor lacks.

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II. PROBLEM SETTING AND PRELIMINARIES

We work with unweighted MaxCut on n -vertex graphs and the depth- p QAOA.

A. MaxCut and the QAOA circuit

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an unweighted graph with $|\mathcal{V}| = n$ and $|\mathcal{E}| = m$. MaxCut asks for a bipartition $z \in \{+1, -1\}^n$ maximizing

$$C(z) = \sum_{(i,j) \in \mathcal{E}} \frac{1}{2}(1 - z_i z_j). \quad (1)$$

The problem is NP-hard [9]. The best known polynomial-time approximation ratio is $\alpha_{\text{GW}} \approx 0.878$ [10], optimal under the Unique Games Conjecture [11, 12].

The depth- p QAOA prepares the state

$$|\gamma, \beta\rangle = e^{-i\beta_p B} e^{-i\gamma_p C} \dots e^{-i\beta_1 B} e^{-i\gamma_1 C} |+\rangle^{\otimes n}, \quad (2)$$

where $C = \sum_{(i,j) \in \mathcal{E}} \frac{1}{2}(I - Z_i Z_j)$ is the cost Hamiltonian and $B = \sum_i X_i$ [1]. The $2p$ angles $\theta = (\gamma, \beta) \in \mathbb{R}^{2p}$ are chosen to maximize $F_{\mathcal{G}}(\theta) = \langle \theta | C | \theta \rangle$.

Two quality measures appear below. The analysis uses the normalized expected objective $F_{\mathcal{G}}(\theta)/C_{\text{max}}$. The experiments report the sampled approximation ratio $r = C(z_{\text{best}})/C_{\text{max}}$, where z_{best} is the best bitstring observed during optimization.

B. Angle concentration and barren plateaus

At fixed depth p on bounded-degree graph families, optimal QAOA angles concentrate: the variance of the optimal θ decreases as $O(1/n)$ [13, 14]. This makes cross-instance parameter transfer possible [3, 15]. The concentration phenomenon is the structural reason that a learned predictor can generalize: without it, the optimal angles for two graphs of the same family could be arbitrarily far apart, and no finite training set would suffice.

Variational circuits can also exhibit barren plateaus, where the gradient variance at random parameters satisfies

$$\text{Var}_{\theta}[\partial_j F(\theta)] \leq O(2^{-n}) \quad (3)$$

for deep or highly entangling ansätze [16–18]. Constant-depth QAOA with a local cost operator avoids this exponential suppression, but the landscape still contains extended flat regions that slow derivative-free solvers. The trust region addresses this difficulty: by confining the search to a neighborhood where the Hessian is bounded away from zero (Theorem 6), it ensures that the optimizer encounters gradient signal throughout its trajectory.

C. Spectral positional encodings

The graph Laplacian $L = D - A$ has eigendecomposition $L = U\Lambda U^\top$ with eigenvalues $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$. The eigenvectors $\{u_k\}$ define spectral positional encodings [7, 8, 19]. The Fiedler value λ_2 [20] captures global connectivity. Because eigenvectors are defined for any graph size, spectral encodings give a size-invariant input representation; this is the mechanism by which the predictor transfers from $n=14$ training graphs to unseen sizes without retraining. Sign ambiguity is resolved by random sign flips during training [8].

D. Graph Isomorphism Networks

The Graph Isomorphism Network (GIN) [6] is as expressive as the Weisfeiler–Leman isomorphism test [21]. Each layer updates

$$h_v^{(l+1)} = \text{MLP}^{(l)}((1 + \epsilon^{(l)}) h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} h_u^{(l)}), \quad (4)$$

with learnable $\epsilon^{(l)}$ and neighbor set $\mathcal{N}(v)$ [22, 23]. A graph-level representation is obtained by permutation-invariant pooling of the node embeddings. The GIN is the strongest first-order message-passing network in the Weisfeiler–Leman hierarchy; a less expressive backbone would collapse distinct spectral signatures that separate the four graph families used in this work.

E. Wasserstein distance for Gaussians

The 2-Wasserstein distance between probability measures P and Q on $\mathbb{R}^d$ is

$$W_2(P, Q) = \left(\inf_{\pi \in \Pi(P, Q)} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\pi(\mathbf{x}, \mathbf{y}) \right)^{1/2}. \quad (5)$$

For diagonal Gaussians this reduces to

$$W_2^2 = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|_2^2 + \|\boldsymbol{\sigma}_1 - \boldsymbol{\sigma}_2\|_2^2, \quad (6)$$

which is differentiable and cheap to evaluate.

III. RELATION TO PRIOR WORK

Parameter transfer for QAOA. Concentration of optimal angles at fixed depth [13, 14] underlies several transfer strategies: fixed-angle schedules [3], interpolation-based warm starts [2], cross-instance transfer [15], warm-starting from related circuits [24, 25], graph-conditioned predictors using GNNs [26] or reinforcement learning [27], Bayesian optimization [28], classical preprocessing [29], training without QPU access [30], unsupervised identification [31], and symmetry-aware search [32]. These methods produce fixed schedules, warm starts, or online search heuristics. Here the

full distribution is predicted before the first circuit query, and the covariance constrains the search. The distinction is not merely architectural: a warm start uses the prediction once and discards it, whereas the trust region enforces the prediction at every iterate. The adaptive budget further couples the model’s uncertainty to the optimizer’s resource consumption, a feature absent from all prior transfer methods.

Learned combinatorial heuristics. Machine learning is used to guide combinatorial optimization [33–38], where objective evaluations are cheap and the metric is time to solution. In the variational quantum setting each circuit evaluation dominates the cost, so the relevant quantity is evaluation count at a given solution quality. The shift in cost metric changes the design problem: runtime-efficient heuristics are no longer sufficient; the algorithm must minimize the number of calls to the quantum device.

Uncertainty quantification. Standard approaches include heteroscedastic regression [39, 40], deep ensembles [41], MC dropout [42], and calibrated prediction [43–46]. Here a single-network Gaussian parameterization suffices; the Wasserstein regularizer and contrastive loss structure the predicted variance without ensembles. The predicted uncertainty sets both the search region and the evaluation budget. A single forward pass replaces the overhead of ensemble disagreement or Monte Carlo sampling, and the Wasserstein regularizer provides the geometric calibration that a standard NLL objective alone cannot supply.

Optimal transport and contrastive learning. The closed-form Wasserstein distance for Gaussians [47] has been used in generative modeling [48, 49] and distributional robustness [50–52]. Here it regularizes a distribution over variational parameters. Contrastive objectives [53–57] organize the embedding so that predicted distributions transfer across graph families. The contrastive loss acts upstream of the prediction heads; it structures the representation by landscape similarity rather than by graph label, which is the mechanism underlying cross-family generalization.

Amortized optimization. Amortized inference replaces per-instance optimization with a learned mapping [58–61]. Existing amortized QAOA methods predict point initializations. Here the prediction is a Gaussian whose covariance constrains the search and whose uncertainty sets the budget. The method is hybrid: learning reshapes per-instance optimization rather than replacing it. The predictor narrows the search; the optimizer finds the solution within the narrowed region. Neither component alone achieves the observed evaluation savings.

IV. GRAPH-CONDITIONED GAUSSIAN PREDICTION AND TRUST-REGION SEARCH

A single forward pass of a GIN produces a Gaussian (μ, Σ) over QAOA angles. Optimization is then restricted to the region and budget that the Gaussian determines. The method has five components (fig. 2): a GIN-based predictor, Wasserstein-regularized training, contrastive metric learning, trust-region-constrained optimization, and uncertainty-guided budget allocation. Each component addresses a distinct failure mode. Removing any one of them either eliminates the uncertainty estimate altogether or degrades it to the point where the trust region provides no advantage over random initialization. The ablation study in section VII confirms this experimentally; the discussion below explains the mechanism.

a. Standing assumptions. The guarantees in section V are local. They assume: (A1) F_G is Lipschitz-continuous on the trust region (Proposition 1); (A2) $\lambda_{\min}(-\nabla^2 F_G(\mu)) \geq \kappa > 0$ (Theorem 6); and (A3) depolarizing noise with per-layer strength $\varepsilon < 1$ (Theorem 18). Assumption (A2) concerns the neighborhood reached by the predictor and does not assert global structure of the QAOA landscape.

A. Gaussian predictor architecture

Given $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, node features are formed by concatenating normalized degree d_v/n with k -dimensional Laplacian spectral positional encodings [7, 8, 19], i.e., the k smallest nontrivial eigenvectors of $L = D - A$, yielding $h_v^{(0)} \in \mathbb{R}^{k+1}$. Eigenvector sign ambiguity is handled by random sign flips during training [8].

A three-layer GIN with hidden dimension $d=64$ and LayerNorm [62] produces node embeddings. A graph-level representation $\mathbf{g} \in \mathbb{R}^{2d}$ is obtained by concatenating mean and max pooling:

$$\mathbf{g} = [\frac{1}{n} \sum_v h_v^{(L)} \parallel \max_v h_v^{(L)}]. \quad (7)$$

Two linear heads produce the Gaussian parameters:

$$\mu = W_\mu \mathbf{g} + b_\mu \in \mathbb{R}^{2p}, \quad (8)$$

$$\log \sigma^2 = \text{clamp}(W_\sigma \mathbf{g} + b_\sigma, -5, 2). \quad (9)$$

The predictive distribution is

$$p(\theta \mid \mathcal{G}) = \mathcal{N}(\mu(\mathcal{G}), \text{diag}(\sigma^2(\mathcal{G}))).$$

The clamp in eq. (9) keeps $\sigma_j \in [e^{-5/2}, e]$. The covariance is diagonal.

The architecture is determined by two requirements. First, the predictor must read the graph: a family-level lookup table or a label-conditioned model cannot generalize to unseen topologies. A GIN is the strongest first-order message-passing network in the Weisfeiler–Leman

hierarchy [6] and distinguishes any pair of graphs that 1-WL distinguishes. A less expressive backbone—GCN or GraphSAGE—collapses distinct spectral signatures onto the same embedding, and the resulting predictions lose resolution across graph families. Second, the predictor must output a distribution, not a point. A point estimate μ alone provides no trust region: the optimizer would have no principled boundary, and the adaptive budget (section IV E) would have no signal. The diagonal Gaussian is the simplest distribution that supplies both a center and a per-coordinate scale.

B. Wasserstein-regularized training

Data. Training uses 240 graphs at $n=14$ (60 per family) and a validation set of 80 graphs. Targets θ_i^* are obtained by multi-restart Nelder–Mead [63] on exact simulation. These are local optima (see section A) and define the regions used for transfer.

Phase 1.

$$\mathcal{L}_1 = \frac{1}{N} \sum_{i=1}^N \|\mu(\mathcal{G}_i) - \theta_i^*\|_2^2 + \lambda_C \mathcal{L}_{\text{CL}}. \quad (10)$$

Phase 2.

$$\mathcal{L}_2 = \mathcal{L}_{\text{NLL}} + \lambda_W \mathcal{L}_{\text{W2}} + \lambda_C \mathcal{L}_{\text{CL}}, \quad (11)$$

The negative log-likelihood is

$$\mathcal{L}_{\text{NLL}} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{2p} \left[\frac{(\mu_j(\mathcal{G}_i) - \theta_{ij}^*)^2}{\sigma_j^2(\mathcal{G}_i)} + \log \sigma_j^2(\mathcal{G}_i) \right]. \quad (12)$$

Wasserstein regularizer. For diagonal Gaussians,

$$W_2^2(P_i, P_j) = \|\mu_i - \mu_j\|_2^2 + \|\sigma_i - \sigma_j\|_2^2.$$

Weights $w_{ij} = \exp(-\|\theta_i^* - \theta_j^*\|_2^2 / 2\tau_W^2)$ encode angle-space similarity:

$$\mathcal{L}_{\text{W2}} = \frac{1}{|\mathcal{B}|^2} \sum_{i \neq j \in \mathcal{B}} w_{ij} W_2^2(P_i, P_j). \quad (13)$$

The negative log-likelihood trains the mean and variance jointly, but it treats each graph in isolation. Two graphs whose ground-truth angles are nearly identical can receive predictive distributions that are far apart in W_2 , and the NLL assigns no penalty for this inconsistency. The Wasserstein term closes this gap: it forces the model to assign similar distributions to graphs with similar optimal angles. Removing it leaves the variance estimates geometrically uncalibrated, and the trust regions no longer reflect landscape similarity across instances.

The two-phase schedule is not a convenience but a necessity. Training the variance head from the first epoch gives the model a degree of freedom it exploits by shrinking σ toward zero, collapsing the Gaussian to a point and

the trust region to a single vector. Phase 1 anchors μ with a squared-error loss; Phase 2 then opens the variance channel under NLL supervision. Reversing or merging the two phases degrades calibration (section I).

C. Contrastive metric learning

A supervised contrastive loss structures the embedding $\mathbf{g}$. For anchor i , define

$$\mathcal{P}(i) = \{j \neq i : \|\theta_i^* - \theta_j^*\|_2 < \delta\}.$$

$$\mathcal{L}_{\text{CL}} = -\frac{1}{|\mathcal{B}|} \sum_i \frac{1}{|\mathcal{P}(i)|} \sum_{j \in \mathcal{P}(i)} \log \frac{e^{\text{sim}(\mathbf{g}_i, \mathbf{g}_j)/\tau_C}}{\sum_{k \neq i} e^{\text{sim}(\mathbf{g}_i, \mathbf{g}_k)/\tau_C}}, \quad (14)$$

The contrastive loss operates on the graph embedding $\mathbf{g}$, not on the output heads. Its role is to organize the representation upstream of the prediction: graphs whose optimal angles are close ($\|\theta_i^* - \theta_j^*\|_2 < \delta$) map to nearby embeddings, and graphs with distant optima are pushed apart. This is complementary to the Wasserstein term, which acts on the predicted distributions downstream. Removing the contrastive loss leaves $\mathbf{g}$ unstructured; the model can still fit training graphs, but transfer to new graph families degrades because the embedding carries no information about landscape similarity.

D. Trust-region QAOA optimization

The predicted covariance defines a constraint on the search.

Definition 1 (Trust region). *Given a predicted Gaussian $\mathcal{N}(\mu, \Sigma)$ for graph $\mathcal{G}$, with Σ symmetric positive definite, and a confidence level $\alpha \in (0, 1)$, the trust region is the Mahalanobis ellipsoid*

$$\mathcal{T}_\alpha(\mathcal{G}) = \{\theta \in \mathbb{R}^{2p} : (\theta - \mu)^\top \Sigma^{-1} (\theta - \mu) \leq \chi_{2p}^2(\alpha)\}, \quad (15)$$

where $\chi_{2p}^2(\alpha)$ is the α -quantile of the χ^2 distribution with $2p$ degrees of freedom.

Optimization is restricted to $\mathcal{T}_\alpha$: (a) candidate angles $\theta_1, \dots, \theta_K \sim \mathcal{N}(\mu, \Sigma)$ are sampled from the corresponding truncated Gaussian restricted to $\mathcal{T}_\alpha$; (b) during Nelder–Mead refinement, any simplex vertex θ' exiting $\mathcal{T}_\alpha$ is projected back:

$$\text{proj}_{\mathcal{T}}(\theta') = \mu + \min\left(1, \frac{\sqrt{\chi_{2p}^2(\alpha)}}{\|\Sigma^{-1/2}(\theta' - \mu)\|_2}\right)(\theta' - \mu). \quad (16)$$

A warm-start method hands μ to the optimizer and allows it to move in any direction; the prediction is used once, at initialization, and has no further effect. The trust region uses the prediction continuously: every iterate is measured against Σ and projected back if it exits

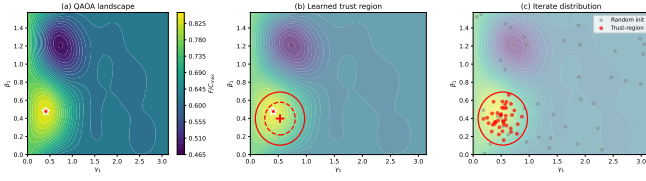


FIG. 1. Trust-region mechanism on an $n=8$ ER instance at $p=2$ (layer-1 slice). (a) QAOA landscape. (b) Learned Mahalanobis regions at 95% and 68%. (c) Random initializations (gray) vs. trust-region samples (red).

$\mathcal{T}_\alpha$. The Mahalanobis ellipsoid is the natural constraint because it is a level set of the predicted Gaussian—the boundary is defined by the model’s own uncertainty, not by an external hyperparameter. Replacing it with a fixed-radius ball discards the per-coordinate calibration that the variance head provides. Unlike warm-start methods [24, 25, 64], which release the optimizer after initialization, the optimizer here is confined to $\mathcal{T}_\alpha$ throughout (fig. 1).

E. Uncertainty-guided adaptive refinement

Define

$$U(\mathcal{G}) = \text{tr}(\Sigma(\mathcal{G}))/2p, \quad z(\mathcal{G}) = \frac{U(\mathcal{G}) - U_{\text{med}}}{U_{\text{iqr}}}.$$

Resource allocation:

$$K(\mathcal{G}) = \text{clamp}(\lfloor 1 + 4\sigma(z) \rfloor, 1, 5), \quad (17)$$

$$T(\mathcal{G}) = \text{clamp}(\lfloor T_{\text{base}}(0.5 + z_+) \rfloor, 5, 2T_{\text{base}}), \quad (18)$$

Higher uncertainty increases both the number of samples and the number of refinement steps. The mapping is monotone and bounded (fig. 11).

Adaptive allocation addresses a problem that no fixed budget can solve. A graph with small predicted uncertainty sits near a well-characterized optimum; one or two samples and a short Nelder–Mead run suffice. A graph with large uncertainty requires broader sampling and more refinement iterations. A fixed budget either over-invests on easy instances or under-invests on hard ones. The mapping from $U(\mathcal{G})$ to (K, T) introduces no free parameters beyond T_{base} ; it is the minimal mechanism that couples the model’s per-instance confidence to the optimizer’s resource consumption. Removing this component and allocating a uniform budget erases the query savings documented in section VII.

Remark 1 (Conformal trust-region radius). *At inference time, the $\chi^2_{2p}(\alpha)$ radius in Step 5 can be replaced by the conformal quantile $\hat{q}_{1-\alpha}$ (proposition 16), yielding a conformalized trust region*

$$\mathcal{T}_{\hat{q}_{1-\alpha}} = \{\theta : (\theta - \mu)^\top \Sigma^{-1}(\theta - \mu) \leq \hat{q}_{1-\alpha}\}$$

Algorithm 1: UQ-QAOA inference

Input: graph $\mathcal{G}$, trained model $f_\theta : \mathcal{G} \mapsto (\mu, \Sigma)$, budget T_{base} , confidence level α .

Output: approximate MaxCut bitstring z^* .

1. Predict $(\mu, \Sigma) \leftarrow f_\theta(\mathcal{G})$.
2. Compute $U \leftarrow \text{tr}(\Sigma)/2p$; $z \leftarrow (U - U_{\text{med}})/U_{\text{iqr}}$.
3. Set sample count $K \leftarrow \text{clamp}(\lfloor 1 + 4\sigma(z) \rfloor, 1, 5)$.
4. Set polish budget $T \leftarrow \text{clamp}(\lfloor T_{\text{base}}(0.5 + z_+) \rfloor, 5, 2T_{\text{base}})$.
5. Construct trust region $\mathcal{T}_\alpha \leftarrow \{\theta : \|\Sigma^{-1/2}(\theta - \mu)\|_2^2 \leq \chi^2_{2p}(\alpha)\}$.
6. Sample $\theta_1, \dots, \theta_K$ from the truncated Gaussian $\mathcal{N}(\mu, \Sigma)$ restricted to $\mathcal{T}_\alpha$.
7. Evaluate QAOA circuit: $\theta^* \leftarrow \arg \max_k F_{\mathcal{G}}(\theta_k)$.
8. Trust-region polish: run T iterations of Nelder–Mead from θ^* , projecting each iterate via eq. (16).
9. Return best bitstring z^* from final QAOA measurement.

FIG. 2. UQ-QAOA inference. Steps 3–4 convert uncertainty into sampling and refinement budgets; Steps 5–8 perform trust-region-constrained optimization.

with the distribution-free, finite-sample guarantee $\mathbb{P}[\theta^* \in \mathcal{T}_{\hat{q}_{1-\alpha}}] \geq 1 - \alpha$. For our validation set ($\hat{q}_{0.90} = 10.8$ vs. $\chi^2_{4}(0.90) = 7.779$), the radius increases by a factor of $(10.8/7.779)^{1/2} \approx 1.178$.

V. GUARANTEES FOR LEARNED TRUST-REGION SEARCH

Each result below is local and conditional on the stated assumptions. The bounds do not assert global optimality; they quantify the worst-case objective loss incurred by confining the optimizer to the predicted ellipsoid and by allocating budget according to the predicted uncertainty. Taken together, the results guarantee that the trust-region mechanism does not introduce additional optimization error beyond what is already present in the landscape near the predicted mean.

Throughout, $\mathcal{G} \sim \mathcal{D}$ denotes a held-out graph except where otherwise stated. The predictor is $f_\theta(\mathcal{G}) = (\mu(\mathcal{G}), \Sigma(\mathcal{G}))$. The reference point $\theta^*(\mathcal{G})$ denotes the local optimum produced by the multi-restart Nelder–Mead procedure used for target generation, or the optimum specified in the corresponding statement. The trust region $\mathcal{T}_\alpha(\mathcal{G})$ is defined in eq. (15). When the radius is not tied to a $\chi^2_{2p}(\alpha)$ quantile, we write $\mathcal{T}_q(\mathcal{G})$.

Expectations and probabilities are taken with respect to the randomness explicitly indicated: draws of $\mathcal{G}$ from $\mathcal{D}$, samples from $\mathcal{N}(\mu(\mathcal{G}), \Sigma(\mathcal{G}))$, or uniform samples on $\mathcal{T}_\alpha(\mathcal{G})$. Constants such as $L_{\mathcal{G},K}$, $L_{\mathcal{G}}$, κ , Λ , and M_3 are deterministic once $\mathcal{G}$ is fixed.

A. Landscape properties

Proposition 1 (Local Lipschitz regularity of the QAOA objective). *For a connected graph $\mathcal{G}$ with m edges, the depth- p QAOA objective $F_{\mathcal{G}} : \mathbb{R}^{2p} \rightarrow \mathbb{R}$ is real analytic. In particular, on any bounded set $K \subset \mathbb{R}^{2p}$,*

$$L_{\mathcal{G},K} := \sup_{\theta \in K} \|\nabla F_{\mathcal{G}}(\theta)\|_2 < \infty. \quad (19)$$

A conservative global bound is $L_{\mathcal{G}} \leq 2m\sqrt{2p}$.

Proof. The QAOA unitary is $U(\theta) = \prod_{\ell=1}^p e^{-i\beta_{\ell}B} e^{-i\gamma_{\ell}C}$, where $B = \sum_v X_v$ and $C = \sum_{(i,j) \in \mathcal{E}} \frac{1}{2}(I - Z_i Z_j)$. Each factor is a matrix exponential of a bounded Hermitian operator, so $U(\theta)$ is an entire function of each angle. The objective $F_{\mathcal{G}}(\theta) = \langle + | U^{\dagger}(\theta) C U(\theta) | + \rangle$ is therefore a real-analytic function on $\mathbb{R}^{2p}$, and $\nabla F_{\mathcal{G}}$ is continuous everywhere. On any compact set K the continuous function $\|\nabla F_{\mathcal{G}}\|_2$ attains its supremum, so $L_{\mathcal{G},K} < \infty$. The mean-value theorem then gives $|F_{\mathcal{G}}(\theta) - F_{\mathcal{G}}(\theta')| \leq L_{\mathcal{G},K} \|\theta - \theta'\|_2$ for all $\theta, \theta' \in K$. For the global bound, note that each partial derivative of $F_{\mathcal{G}}$ is an expectation value of a commutator. For the phase-separation angle γ_{ℓ} , $\partial_{\gamma_{\ell}} F_{\mathcal{G}} = -i \langle \theta | [C_{\ell}, U_{\ell}^{\dagger} C U_{\ell}] | \theta \rangle$ where C_{ℓ} is the cost operator acting in layer ℓ . Since C has operator norm $\|C\|_{\text{op}} \leq m$ (each edge contributes at most 1), the commutator has norm at most $2m$, giving $|\partial_{\gamma_{\ell}} F_{\mathcal{G}}| \leq m$. Similarly, $|\partial_{\beta_{\ell}} F_{\mathcal{G}}| \leq n$ since $\|B\|_{\text{op}} = n$. The full gradient satisfies $\|\nabla F_{\mathcal{G}}\|_2^2 = \sum_{\ell=1}^p [(\partial_{\gamma_{\ell}} F)^2 + (\partial_{\beta_{\ell}} F)^2] \leq p(m^2 + n^2)$. For connected graphs, $m \geq n - 1$ so $m^2 + n^2 \leq 2m^2 + 2m + 1 \leq 4m^2$ for $m \geq 2$. Hence $\|\nabla F_{\mathcal{G}}\|_2 \leq 2m\sqrt{p} \leq 2m\sqrt{2p}$, which establishes the global bound. $\square$

The Lipschitz constant bounds the worst-case objective change per unit displacement in angle space. For practical instances, the local constant $L_{\mathcal{G},K}$ on the trust region is much smaller than the global bound $2m\sqrt{2p}$; the global bound serves only to confirm that the constant is finite and computable.

B. Trust-region guarantees

Lemma 2 (Validity of the Mahalanobis projection). *Let $\mu \in \mathbb{R}^{2p}$, $\Sigma \succ 0$, and $q > 0$. For any $\theta' \in \mathbb{R}^{2p}$, the map*

$$\text{proj}_{\mathcal{T}}(\theta') = \mu + \min\left(1, \frac{\sqrt{q}}{\|\Sigma^{-1/2}(\theta' - \mu)\|_2}\right)(\theta' - \mu)$$

satisfies:

- (i) $\text{proj}_{\mathcal{T}}(\theta') \in \mathcal{T}_q$ for every $\theta' \in \mathbb{R}^{2p}$.
- (ii) If $\theta' \in \mathcal{T}_q$, then $\text{proj}_{\mathcal{T}}(\theta') = \theta'$ (idempotency).
- (iii) $\text{proj}_{\mathcal{T}}$ is the nearest point in $\mathcal{T}_q$ under the Mahalanobis norm $\|\cdot\|_{\Sigma^{-1}}$.

Proof. Write $d = \theta' - \mu$ and $s = \|\Sigma^{-1/2}d\|_2$. The projected point is $\theta_{\text{pr}} = \mu + \min(1, \sqrt{q}/s)d$.

(i) If $s \leq \sqrt{q}$, then $\min(1, \sqrt{q}/s) = 1$ and $\theta_{\text{pr}} = \theta'$, which satisfies the constraint $s^2 \leq q$. If $s > \sqrt{q}$,

$$\|\Sigma^{-1/2}(\theta_{\text{pr}} - \mu)\|_2 = \frac{\sqrt{q}}{s} s = \sqrt{q},$$

so θ_{pr} lies on the boundary of $\mathcal{T}_q$.

(ii) Immediate from the first case above.

(iii) For $s > \sqrt{q}$, the projection scales d to the boundary along the direction d/s . The Mahalanobis ball $\mathcal{T}_q$ is convex, and radial scaling toward the center is the Euclidean projection in the whitened coordinates $u = \Sigma^{-1/2}(\theta - \mu)$, which is the nearest-point projection onto the ball $\|u\|_2 \leq \sqrt{q}$. $\square$

Theorem 3 (Lower bound on the expected objective within the trust region). *Let $\mathcal{T}_{\alpha}(\mathcal{G})$ be defined by eq. (15). Suppose $F_{\mathcal{G}}(\mu) \geq r^*C_{\text{max}}$. Let Θ be supported on $\mathcal{T}_{\alpha}(\mathcal{G})$. Then*

$$\mathbb{E}[F_{\mathcal{G}}(\Theta)] \geq r^*C_{\text{max}} - L_{\mathcal{G}} \sqrt{\chi_{2p}^2(\alpha) \cdot \|\Sigma\|_{\text{op}}}, \quad (20)$$

Proof. For $\theta \in \mathcal{T}_{\alpha}$, one has

$$\|\theta - \mu\|_2^2 \leq \|\Sigma\|_{\text{op}} \cdot (\theta - \mu)^{\top} \Sigma^{-1} (\theta - \mu) \leq \|\Sigma\|_{\text{op}} \cdot \chi_{2p}^2(\alpha).$$

Hence

$$\|\theta - \mu\|_2 \leq \sqrt{\chi_{2p}^2(\alpha) \|\Sigma\|_{\text{op}}}.$$

By Proposition 1,

$$|F_{\mathcal{G}}(\theta) - F_{\mathcal{G}}(\mu)| \leq L_{\mathcal{G}} \sqrt{\chi_{2p}^2(\alpha) \|\Sigma\|_{\text{op}}}.$$

Applying the assumption $F_{\mathcal{G}}(\mu) \geq r^*C_{\text{max}}$ and taking expectation yields the claim. $\square$

Corollary 4 (Expected objective under variance confinement). *Under the conditions of Theorem 3, if $\sigma_{\text{max}}^2 \leq (r^*C_{\text{max}})^2 / (4L_{\mathcal{G}}^2 \chi_{2p}^2(\alpha))$, then $\mathbb{E}[F_{\mathcal{G}}(\Theta)] \geq \frac{1}{2}r^*C_{\text{max}}$.*

Proof. Set $\epsilon = \frac{1}{2}r^*C_{\text{max}}$ in Theorem 3. This gives the sufficient condition $\sigma_{\text{max}}^2 \leq (r^*C_{\text{max}})^2 / (4L_{\mathcal{G}}^2 \chi_{2p}^2(\alpha))$. The conclusion follows immediately. $\square$

The corollary states a concrete variance threshold: if the predicted variance is small enough relative to the Lipschitz constant and the objective value at the mean, the expected objective inside the trust region remains at least half of the best achievable. The threshold is inversely quadratic in the Lipschitz constant, so tighter local Lipschitz estimates yield proportionally larger allowable variance.

Proposition 5 (Second-order local improvement within the trust region). *Assume $F_{\mathcal{G}}$ is twice continuously differentiable near μ with $\lambda_{\min}(-\nabla^2 F_{\mathcal{G}}(\mu)) \geq \kappa > 0$ (i.e.,*

$\boldsymbol{\mu}$ is a strict local maximizer). If the trust region satisfies $\sqrt{\chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}}} \leq \kappa/M_3$, where M_3 bounds the third derivatives on $\mathcal{T}_\alpha$, then for every $\boldsymbol{\theta} \in \mathcal{T}_\alpha$,

$$F_G(\boldsymbol{\theta}) \geq F_G(\boldsymbol{\mu}) - \frac{1}{2}\Lambda \chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}} - \frac{1}{6}M_3 [\chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}}]^{3/2}, \quad (21)$$

where $\Lambda = \|\nabla^2 F_G(\boldsymbol{\mu})\|_{\text{op}}$. When the trust region is small the leading penalty is quadratic in $\sigma_{\max}$ rather than linear.

Proof. By Taylor expansion around $\boldsymbol{\mu}$ (a critical point, so $\nabla F_G(\boldsymbol{\mu}) = 0$ for a local maximizer):

$$F_G(\boldsymbol{\theta}) = F_G(\boldsymbol{\mu}) + \frac{1}{2}\Delta^\top \nabla^2 F_G(\boldsymbol{\mu}) \Delta + O(\|\Delta\|_2^3),$$

where $\Delta = \boldsymbol{\theta} - \boldsymbol{\mu}$. Since $\boldsymbol{\mu}$ is a local maximizer, $\nabla^2 F_G(\boldsymbol{\mu}) \preceq 0$, so $\Delta^\top \nabla^2 F_G(\boldsymbol{\mu}) \Delta \geq -\Lambda \|\Delta\|_2^2$. The cubic remainder is bounded by $\frac{1}{6}M_3(2p)^{3/2}\|\Delta\|_2^3 \leq \frac{1}{6}M_3\|\Delta\|_2^3$ (absorbing dimension factors into M_3). For $\boldsymbol{\theta} \in \mathcal{T}_\alpha$, $\|\Delta\|_2^2 \leq \chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}}$ (as in Theorem 3). Substituting:

$$\begin{aligned} F_G(\boldsymbol{\theta}) &\geq F_G(\boldsymbol{\mu}) - \frac{1}{2}\Lambda \|\Delta\|_2^2 - \frac{1}{6}M_3 \|\Delta\|_2^3 \\ &\geq F_G(\boldsymbol{\mu}) - \frac{1}{2}\Lambda \chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}} \\ &\quad - \frac{1}{6}M_3 [\chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}}]^{3/2}. \quad \square \end{aligned}$$

C. Gradient anti-concentration

Theorem 6 (Local gradient anti-concentration in the trust region). Assume F_G is three times continuously differentiable near $\mathcal{T}_\alpha(\mathcal{G})$, $\lambda_{\min}(-\nabla^2 F_G(\boldsymbol{\mu})) \geq \kappa > 0$, and third derivatives are bounded by M_3 . Then for $\boldsymbol{\theta} \sim \text{Unif}(\mathcal{T}_\alpha)$,

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta} \sim \text{Unif}(\mathcal{T}_\alpha)}[\partial_j F_G(\boldsymbol{\theta})] &\geq \frac{\kappa^2 \sigma_{\min}^2 \chi_{2p}^2(\alpha)}{2p+2} \\ &\quad - C_{p,\alpha} (\Lambda M_3 \sigma_{\max}^3 + M_3^2 \sigma_{\max}^4), \end{aligned} \quad (22)$$

where $\Lambda := \|\nabla^2 F_G(\boldsymbol{\mu})\|_{\text{op}}$, $\sigma_{\min} = \min_j \sigma_j$, and $\sigma_{\max} = \max_j \sigma_j$.

Proof. See section H. $\square$

The leading term is strictly positive whenever the Hessian at $\boldsymbol{\mu}$ has a nonzero eigenvalue and $\sigma_{\min} > 0$. The remainder becomes negligible when $\sigma_{\max}$ is small. This is the central guarantee that the trust region avoids the barren-plateau phenomenon locally: inside the ellipsoid, the gradient variance is bounded below by a quantity proportional to $\kappa^2 \sigma_{\min}^2$, regardless of the circuit depth. The bound applies to derivative-free methods as well, because a positive gradient variance implies that the objective varies sufficiently across the trust region for a simplex-based solver to make progress.

D. Algorithmic guarantees

Proposition 7 (Feasibility and monotonic best-so-far value under projection). Projected optimization within $\mathcal{T}_\alpha$ produces feasible iterates and a nondecreasing best-so-far objective. The final iterate dominates the initial sampled candidates. Let $\{\boldsymbol{\theta}_t\}_{t=0}^T$ be the feasible iterates generated by projected Nelder–Mead within $\mathcal{T}_\alpha$, define $b_t := \max_{0 \leq s \leq t} F_G(\boldsymbol{\theta}_s)$, and let $\hat{\boldsymbol{\theta}}_t$ denote any feasible iterate attaining b_t . Then:

(i) $\boldsymbol{\theta}_t \in \mathcal{T}_\alpha$ for all t .

(ii) b_t is non-decreasing in t .

(iii) $F_G(\hat{\boldsymbol{\theta}}_T) \geq \max_k F_G(\boldsymbol{\theta}_k)$ where $\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K$ are the initial feasible samples used to seed the search.

Proof. (i) The projection operator eq. (16) maps every point to $\mathcal{T}_\alpha$: if $\boldsymbol{\theta}' \in \mathcal{T}_\alpha$, the projection is the identity; otherwise, it rescales $\boldsymbol{\theta}' - \boldsymbol{\mu}$ to the boundary of $\mathcal{T}_\alpha$. (ii) Monotonicity is immediate from the definition $b_t = \max_{0 \leq s \leq t} F_G(\boldsymbol{\theta}_s)$: the index set on which the maximum is taken only enlarges with t . (iii) Because the initial sampled points are among the feasible points seen by the algorithm before refinement begins, $b_T \geq \max_{1 \leq k \leq K} F_G(\boldsymbol{\theta}_k)$. Since $\hat{\boldsymbol{\theta}}_T$ attains b_T , the claim follows. $\square$

Corollary 8 (Output guarantee under trust-region capture). Let $\boldsymbol{\theta}_{\text{out}}$ denote the best feasible angle vector returned by Algorithm 2, and let z_{out} be the final bitstring sampled from the QAOA circuit at $\boldsymbol{\theta}_{\text{out}}$. Fix a trust region $\mathcal{T}_q(\mathcal{G})$ and assume the reference local optimum $\boldsymbol{\theta}^*(\mathcal{G})$ belongs to that region. Then

$$F_G(\boldsymbol{\theta}_{\text{out}}) \geq F_G(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_G \sqrt{q \|\boldsymbol{\Sigma}(\mathcal{G})\|_{\text{op}}}, \quad (23)$$

and therefore

$$\begin{aligned} \mathbb{E}[C(z_{\text{out}}) \mid \boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})] &\geq F_G(\boldsymbol{\theta}^*(\mathcal{G})) \\ &\quad - 2L_G \sqrt{q \|\boldsymbol{\Sigma}(\mathcal{G})\|_{\text{op}}}. \end{aligned} \quad (24)$$

If, in addition, the trust region has capture probability $p_{\text{cap}}(\mathcal{G})$, then since MaxCut costs are nonnegative,

$$\mathbb{E}[C(z_{\text{out}})] \geq p_{\text{cap}}(\mathcal{G}) \left[F_G(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_G \sqrt{q \|\boldsymbol{\Sigma}(\mathcal{G})\|_{\text{op}}} \right]. \quad (25)$$

For the algorithmic χ^2 region one may take $p_{\text{cap}} \geq \alpha - \eta$ from Proposition 17; for the conformalized region one has $p_{\text{cap}} \geq 1 - \alpha$.

Proof. Under the capture event, every sampled candidate and every projected iterate lies in $\mathcal{T}_q(\mathcal{G})$. Proposition 17 therefore gives the pointwise bound

$$F_G(\boldsymbol{\theta}) \geq F_G(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_G \sqrt{q \|\boldsymbol{\Sigma}(\mathcal{G})\|_{\text{op}}} \quad \forall \boldsymbol{\theta} \in \mathcal{T}_q(\mathcal{G}).$$

Since Algorithm 2 returns an angle vector inside that region, eq. (23) follows. By definition of the QAOA objective, $\mathbb{E}[C(z_{\text{out}}) \mid \theta_{\text{out}}] = F_{\mathcal{G}}(\theta_{\text{out}})$, which gives eq. (24). The unconditional bound eq. (25) follows by conditioning on the capture event and using $C(z_{\text{out}}) \geq 0$ for Max-Cut. $\square$

E. Sampling guarantees

Theorem 9 (Best-of- K guarantee). *Let $\theta_1, \dots, \theta_K \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \Sigma)$. Then*

$$\mathbb{E}[\max_k F_{\mathcal{G}}(\theta_k)] \geq F_{\mathcal{G}}(\mu) - \frac{L_{\mathcal{G}} \sqrt{\text{tr}(\Sigma)}}{\sqrt{K}}. \quad (26)$$

Proof. See section C. $\square$

Lemma 10 (Trust-region volume ratio). *For a diagonal covariance $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_{2p}^2)$ with confidence $\alpha=0.95$ and $p=2$, the trust-region volume relative to the unconstrained search hypercube $[-\pi, \pi]^{2p}$ is*

$$\frac{\text{Vol}(\mathcal{T}_{0.95})}{\text{Vol}([-\pi, \pi]^{2p})} = \frac{\pi^p [\chi_4^2(0.95)]^p}{\Gamma(p+1) (2\pi)^{2p}} \prod_{j=1}^4 \sigma_j. \quad (27)$$

If one uses the isotropic proxy $\sigma_j \equiv \bar{\sigma} = 0.15$, then eq. (27) gives $0.2850 \bar{\sigma}^4 \approx 1.44 \times 10^{-4}$.

Proof. The trust region is a 4-dimensional ellipsoid with half-lengths $\sigma_j \sqrt{\chi_4^2(0.95)}$. Its volume is $\frac{\pi^2}{2} [\chi_4^2(0.95)]^2 \prod_j \sigma_j$, and the hypercube volume is $(2\pi)^4$. The ratio follows by direct computation with $\chi_4^2(0.95) \approx 9.49$. $\square$

Each additional sample improves the expected initial value at rate $1/\sqrt{K}$. The trust-region volume at $p=2$ and $\alpha=0.95$ is roughly 10^{-4} of the full search space. The volume reduction explains the efficiency gain in geometric terms: the optimizer samples from a region four orders of magnitude smaller than the unconstrained hypercube, so each evaluation is far more likely to land near the optimum.

F. Generalization and calibration

Proposition 11 (Wasserstein continuity in a WL-compatible pseudometric). *If the GIN encoder ϕ is L_{enc} -Lipschitz with respect to a WL-compatible graph pseudometric d_{WL} and the prediction heads are L_{head} -Lipschitz, then:*

$$W_2(f_{\theta}(\mathcal{G}), f_{\theta}(\mathcal{G}')) \leq \sqrt{2} L_{\text{head}} L_{\text{enc}} d_{\text{WL}}(\mathcal{G}, \mathcal{G}'), \quad (28)$$

where $d_{\text{WL}}(\mathcal{G}, \mathcal{G}') = 0$ is allowed for non-isomorphic graphs that are indistinguishable by the relevant WL statistics.

Proof. See section E. $\square$

Theorem 12 (Adaptive-allocation guarantee under calibration and monotone difficulty). *Assume approximate calibration and the regret envelope eq. (30). Suppose the predicted uncertainty satisfies η -approximate calibration:*

$$|\mathbb{P}[\|\theta^* - \mu\|_{\Sigma^{-1}} \leq r] - F_{\chi_{2p}^2}(r^2)| \leq \eta \quad \forall r > 0, \quad (29)$$

where $F_{\chi_{2p}^2}$ is the χ_{2p}^2 CDF. Assume the residual optimization regret satisfies the instance-wise envelope

$$R_{\pi}(\mathcal{G}) \leq c_0 + c_1 \frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_{\pi}(\mathcal{G})}} \quad (30)$$

where $U(\mathcal{G}) = \text{tr}(\Sigma(\mathcal{G}))/2p$ and $c_0, c_1 \geq 0$ are independent of π . Define population regret by

$$\text{Regret}(\pi) := \mathbb{E}_{\mathcal{G} \sim \mathcal{D}}[R_{\pi}(\mathcal{G})]. \quad (31)$$

Let $\bar{K} := \mathbb{E}_{\mathcal{D}}[K_{\text{adapt}}(\mathcal{G})]$, and let π_{unif} denote the mean-budget-matched uniform comparator with $K_{\text{unif}}(\mathcal{G}) \equiv \bar{K}$. Assume further that the adaptive allocation rule satisfies the envelope-dominance condition

$$\mathbb{E}_{\mathcal{D}} \left[\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_{\text{adapt}}(\mathcal{G})}} \right] \leq \frac{1}{\sqrt{\bar{K}}} \mathbb{E}_{\mathcal{D}} [\sqrt{U(\mathcal{G}) + \eta}]. \quad (32)$$

Then the adaptive allocation strategy π_{adapt} (Eqs. 17–18) satisfies

$$\begin{aligned} \text{Regret}(\pi_{\text{adapt}}) &\leq c_0 + c_1 \mathbb{E}_{\mathcal{D}} \left[\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_{\text{adapt}}(\mathcal{G})}} \right] \\ &\leq c_0 + \frac{c_1}{\sqrt{\bar{K}}} \mathbb{E}_{\mathcal{D}} [\sqrt{U(\mathcal{G}) + \eta}], \end{aligned} \quad (33)$$

$$\mathbb{E}[B_{\pi_{\text{adapt}}}] = \alpha \mathbb{E}[B_{\pi_{\text{unif}}}], \quad (34)$$

where B_{π} is the total evaluation budget under policy π and $\alpha = \mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{adapt}}}] / \mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{unif}}}]$ depends on the deployment distribution of predicted uncertainties. Thus $\alpha < 1$ is not a consequence of calibration alone; it depends on the deployment distribution together with the implemented allocation rule.

Proof. See section D. $\square$

Proposition 13 (Monotonicity of the implemented allocation rule). *Let $z(\mathcal{G}) = (U(\mathcal{G}) - U_{\text{med}})/U_{\text{iqr}}$, and define K_{adapt} and T_{adapt} by eqs. (17) and (18). Then both allocations are nondecreasing functions of U , and one has the explicit thresholds*

$$K_{\text{adapt}}(\mathcal{G}) = \begin{cases} 1, & z(\mathcal{G}) < -\log 3, \\ 2, & -\log 3 \leq z(\mathcal{G}) < 0, \\ 3, & 0 \leq z(\mathcal{G}) < \log 3, \\ 4, & z(\mathcal{G}) \geq \log 3. \end{cases} \quad (35)$$

If $T_{\text{base}} \geq 10$, then $T_{\text{adapt}}(\mathcal{G}) = \lfloor T_{\text{base}}/2 \rfloor$ for $z(\mathcal{G}) \leq 0$, and T_{adapt} increases linearly in $z(\mathcal{G})$ until it reaches the

cap $2T_{\text{base}}$. Thus the implemented policy reallocates computation monotonically from the low- U region to the high- U region, which is the ordering premise behind eq. (32). The same ordering is supported empirically by fig. 9 and table I: larger predicted uncertainty is associated with harder instances, and exploiting that ordering lowers the total query count.

Proof. The sigmoid is strictly increasing, so $\lfloor 1 + 4\sigma(z) \rfloor$ is nondecreasing in z . The thresholds follow from the identities $\sigma(z) = 1/4$ at $z = -\log 3$, $\sigma(z) = 1/2$ at $z = 0$, and $\sigma(z) = 3/4$ at $z = \log 3$. Because $\sigma(z) < 1$ for finite z , the implemented rule takes the values 1, 2, 3, 4 on the four regimes in eq. (35). The monotonicity of T_{adapt} follows in the same way from the monotonicity of $z_+ = \max(z, 0)$ and the clamp map. If $z \leq 0$, then $z_+ = 0$ and $T_{\text{adapt}} = \text{clamp}(\lfloor T_{\text{base}}/2 \rfloor, 5, 2T_{\text{base}}) = \lfloor T_{\text{base}}/2 \rfloor$ whenever $T_{\text{base}} \geq 10$. The empirical statements are exactly those reported in fig. 9 and table I. $\square$

Theorem 14 (Spectral-norm generalization guarantee). *Let f_θ be the trained GIN predictor with L layers, hidden dimension d , and per-layer spectral norms $s_\ell = \|W_\ell\|_\sigma$. Let $B_x = \max_G \|\mathbf{h}^{(0)}\|_2$ be the bound on input features. Assume the induced loss class $\mathcal{F} = \{W_2^2(f_\theta(\cdot), \delta_{\theta^*}(\cdot))\}$ is uniformly bounded by M and satisfies the empirical Rademacher complexity estimate*

$$\mathfrak{R}_N(\mathcal{F}) \leq \frac{2B_x \prod_{\ell=1}^L s_\ell \sqrt{2L \log(2d)}}{\sqrt{N}}.$$

Given N i.i.d. training graphs, the population Wasserstein risk $\mathcal{R}(f_\theta) := \mathbb{E}_{G \sim \mathcal{D}}[W_2^2(f_\theta(G), \delta_{\theta^}(G))]$ satisfies, with probability $\geq 1 - \delta$ over the training set:*

$$\begin{aligned} \mathcal{R}(f_\theta) \leq \hat{\mathcal{R}}_N + \frac{4B_x \prod_{\ell=1}^L s_\ell}{\sqrt{N}} \sqrt{2L \log(2d)} \\ + 3M \sqrt{\frac{\log(2/\delta)}{2N}}, \end{aligned} \quad (36)$$

where $\hat{\mathcal{R}}_N$ is the empirical Wasserstein risk and M is the loss bound. Here $\delta_{\theta^*}(G)$ denotes the Dirac point mass at the target angle vector. The complexity term scales with $\prod_\ell s_\ell \sqrt{L \log d/N}$ rather than with the total parameter count [65–67].

Proof. See section F. $\square$

Corollary 15 (Sample-complexity consequence). *Under the conditions of Theorem 14, to achieve population risk within ϵ of the empirical risk with probability $\geq 1 - \delta$, it suffices to train on*

$$N \geq O\left(\frac{B_x^2 (\prod_\ell s_\ell)^2 L \log d}{\epsilon^2} + \frac{M^2 \log(1/\delta)}{\epsilon^2}\right) \quad (37)$$

graphs.

Proof. Set the generalization gap

$$\frac{4B_x \prod s_\ell \sqrt{2L \log(2d)}}{\sqrt{N}} + 3M \sqrt{\frac{\log(2/\delta)}{2N}} \leq \epsilon$$

and solve for N . A sufficient condition is that both terms are bounded by $\epsilon/2$, which holds when $N \geq \frac{128B_x^2 (\prod s_\ell)^2 L \log(2d)}{\epsilon^2}$ and $N \geq \frac{18M^2 \log(2/\delta)}{\epsilon^2}$ respectively. $\square$

The generalization gap scales with $\prod_\ell s_\ell \sqrt{L \log d/N}$ rather than the total parameter count. The conformal guarantee below compensates for looseness at small N by providing a non-vacuous, distribution-free statement. The two guarantees are complementary: the spectral-norm bound governs the average-case risk and improves with more data; the conformal bound gives a worst-case, finite-sample statement that holds at any N without distributional assumptions.

G. Conformal coverage

Proposition 16 (Finite-sample conformal coverage). *Let $\{(\mathcal{G}_i, \theta_i^*)\}_{i=1}^M$ be an exchangeable validation set. Define the nonconformity score*

$$s(\mathcal{G}) = (\theta^*(\mathcal{G}) - \mu(\mathcal{G}))^\top \Sigma^{-1}(\mathcal{G}) (\theta^*(\mathcal{G}) - \mu(\mathcal{G})).$$

Let $k = \lceil (M+1)(1-\alpha) \rceil$, and let $\hat{q}_{1-\alpha}$ be the k -th order statistic of $\{s(\mathcal{G}_i)\}_{i=1}^M$. Then for any new exchangeable test instance $\mathcal{G}_{M+1}$:

$$\mathbb{P}[\theta^*(\mathcal{G}_{M+1}) \in \mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G}_{M+1})] \geq 1 - \alpha, \quad (38)$$

where

$$\mathcal{T}_q(\mathcal{G}) = \{\theta : (\theta - \mu)^\top \Sigma^{-1}(\theta - \mu) \leq q\}$$

[68–70]. The coverage statement applies to the target optima θ_i^* used throughout the paper.

Proof. By the exchangeability of $\{(\mathcal{G}_i, \theta_i^*)\}_{i=1}^{M+1}$, the rank of $s(\mathcal{G}_{M+1})$ among $\{s(\mathcal{G}_1), \dots, s(\mathcal{G}_{M+1})\}$ is uniformly distributed on $\{1, \dots, M+1\}$. Therefore $\mathbb{P}[s(\mathcal{G}_{M+1}) \leq \hat{q}_{1-\alpha}] = \mathbb{P}[\text{rank}(s(\mathcal{G}_{M+1})) \leq k] \geq k/(M+1) \geq 1 - \alpha$. Since $s(\mathcal{G}) \leq q \Leftrightarrow \theta^* \in \mathcal{T}_q(\mathcal{G})$, the result follows [68, 70]. $\square$

Proposition 17 (Trust-region capture and local objective control under calibration). *Let $\theta^*(\mathcal{G})$ denote the reference local optimum for graph $\mathcal{G}$, and let Θ be any random variable supported on a trust region centered at the learned mean $\mu(\mathcal{G})$.*

(a) Algorithmic χ^2 region. *If the Mahalanobis score satisfies the η -approximate calibration condition in eq. (29), then for the trust region $\mathcal{T}_\alpha(\mathcal{G})$ from definition 1,*

$$\mathbb{P}[\theta^*(\mathcal{G}) \in \mathcal{T}_\alpha(\mathcal{G})] \geq \alpha - \eta. \quad (39)$$

(b) Conformal region. For the conformalized trust region $\mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G})$ from Proposition 16,

$$\mathbb{P}[\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_{\hat{q}_{1-\alpha}}(\mathcal{G})] \geq 1 - \alpha. \quad (40)$$

(c) Local objective control after capture. Let

$$R(\mathcal{G}; q) := \sqrt{q \|\boldsymbol{\Sigma}(\mathcal{G})\|_{\text{op}}}. \quad (41)$$

On the event $\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})$, every $\boldsymbol{\theta} \in \mathcal{T}_q(\mathcal{G})$ satisfies

$$\|\boldsymbol{\theta} - \boldsymbol{\theta}^*(\mathcal{G})\|_2 \leq 2R(\mathcal{G}; q), \quad (42)$$

and therefore, by Lipschitz continuity,

$$F_{\mathcal{G}}(\boldsymbol{\theta}) \geq F_{\mathcal{G}}(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_{\mathcal{G}}R(\mathcal{G}; q). \quad (43)$$

Consequently,

$$\mathbb{E}[F_{\mathcal{G}}(\boldsymbol{\theta}) \mid \boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_q(\mathcal{G})] \geq F_{\mathcal{G}}(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_{\mathcal{G}}R(\mathcal{G}; q). \quad (44)$$

Proof. For part (a), apply eq. (29) with $r = \sqrt{\chi_{2p}^2(\alpha)}$. Since $F_{\chi_{2p}^2}(\chi_{2p}^2(\alpha)) = \alpha$,

$$\begin{aligned} \mathbb{P}[\boldsymbol{\theta}^*(\mathcal{G}) \in \mathcal{T}_{\alpha}(\mathcal{G})] &= \mathbb{P}[\|\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}(\mathcal{G})\|_{\boldsymbol{\Sigma}^{-1}}^2 \leq \chi_{2p}^2(\alpha)] \\ &\geq \alpha - \eta. \end{aligned}$$

Part (b) is exactly Proposition 16.

For part (c), if both $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^*(\mathcal{G})$ belong to $\mathcal{T}_q(\mathcal{G})$, then by the Euclidean radius bound associated with eq. (15),

$$\|\boldsymbol{\theta} - \boldsymbol{\mu}\|_2 \leq R(\mathcal{G}; q), \quad \|\boldsymbol{\theta}^*(\mathcal{G}) - \boldsymbol{\mu}\|_2 \leq R(\mathcal{G}; q).$$

The triangle inequality gives eq. (42). Applying Proposition 1 around the reference local optimum then yields

$$\begin{aligned} F_{\mathcal{G}}(\boldsymbol{\theta}) &\geq F_{\mathcal{G}}(\boldsymbol{\theta}^*(\mathcal{G})) - L_{\mathcal{G}}\|\boldsymbol{\theta} - \boldsymbol{\theta}^*(\mathcal{G})\|_2 \\ &\geq F_{\mathcal{G}}(\boldsymbol{\theta}^*(\mathcal{G})) - 2L_{\mathcal{G}}R(\mathcal{G}; q), \end{aligned}$$

which is eq. (43). Taking conditional expectation over any $\boldsymbol{\theta}$ supported on $\mathcal{T}_q(\mathcal{G})$ gives eq. (44). $\square$

Remark 2 (Finite-scale sharpness). The global Lipschitz prefactor $L_{\mathcal{G}} = 2m\sqrt{2p}$ makes Theorem 3 vacuous at $n=14$. Proposition 5 is tighter when $\boldsymbol{\mu}$ is near a local maximum. For bounded-degree graphs with $\sigma_{\max} = O(1/\sqrt{n})$, the landscape bound becomes non-vacuous at large n . The conformal guarantee (Proposition 16) already provides a non-vacuous coverage bound at the experimental scale.

Remark 3 (Numerical instantiation at the operating scale). At $n=14$ ($m \approx 46$, $\sigma_{\max} \approx 0.15$, $\chi_4^2(0.95) = 9.49$), the second-order penalty (Proposition 5) is $\frac{1}{2}\Lambda \cdot 0.214 \approx 0.107\Lambda$. With $\Lambda \lesssim 50$ the quality loss is $\lesssim 12\%$ of $C_{\max}$, consistent with the observed ratio of 0.956. The first-order Lipschitz bound gives $\sim 85\%$, exceeding $C_{\max} \approx 46$ and is therefore vacuous at this scale.

H. Noise robustness

The preceding results hold for noiseless circuits. The following theorem addresses depolarizing noise.

Theorem 18 (Preservation of expected ordering under depolarizing noise). Consider a global depolarizing noise model where the noisy QAOA state after depth p satisfies

$$\rho^{\varepsilon} = (1 - \nu) |\boldsymbol{\theta}\rangle\langle\boldsymbol{\theta}| + \nu \frac{I}{2^n}, \quad (45)$$

with noise strength $\nu = 1 - (1 - \varepsilon)^{2p}$, $\varepsilon \in [0, 1]$. The noisy objective is

$$F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\theta}) = (1 - \nu) F_{\mathcal{G}}(\boldsymbol{\theta}) + \nu \bar{C}, \quad (46)$$

where $\bar{C} = \text{tr}(C)/2^n = m/2$ is the expected MaxCut cost at the maximally mixed state. Let $\boldsymbol{\Theta}_{\text{TR}}$ be any random variable supported on $\mathcal{T}_{\alpha}(\mathcal{G})$ almost surely, and let $\boldsymbol{\Theta}_{\text{rand}}$ denote a random initialization baseline. Assume also that the predicted mean satisfies $F_{\mathcal{G}}(\boldsymbol{\mu}) \geq r^* C_{\max}$ for some $r^* > 0$. Then:

(a) Expected noisy landscape bound.

$$\begin{aligned} \mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{TR}})] &\geq (1 - \nu) [r^* C_{\max} - L_{\mathcal{G}} \sqrt{\chi_{2p}^2(\alpha) \|\boldsymbol{\Sigma}\|_{\text{op}}}] \\ &\quad + \nu \bar{C}. \end{aligned} \quad (47)$$

(b) Preserved expected ordering.

$$\begin{aligned} \mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{TR}})] - \mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{rand}})] &= (1 - \nu) \left(\mathbb{E}[F_{\mathcal{G}}(\boldsymbol{\Theta}_{\text{TR}})] \right. \\ &\quad \left. - \mathbb{E}[F_{\mathcal{G}}(\boldsymbol{\Theta}_{\text{rand}})] \right). \end{aligned} \quad (48)$$

Hence any noiseless expected advantage is preserved for $\varepsilon < 1$.

(c) Preserved relative expected advantage.

$$\frac{\mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{rand}})] - \bar{C}} = \frac{\mathbb{E}[F_{\mathcal{G}}(\boldsymbol{\Theta}_{\text{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}(\boldsymbol{\Theta}_{\text{rand}})] - \bar{C}}, \quad (49)$$

The ratio statement in part (c) is understood whenever the denominator $\mathbb{E}[F_{\mathcal{G}}^{\varepsilon}(\boldsymbol{\Theta}_{\text{rand}})] - \bar{C}$ is nonzero.

Proof. See section G. $\square$

The expected ordering between trust-region search and random initialization is preserved for any $\varepsilon < 1$, and the relative excess objective is invariant under noise. This invariance is a consequence of the affine structure of the depolarizing channel: the noise contracts all objectives toward the same fixed point $\bar{C}$, so any advantage in the noiseless setting survives multiplication by $(1 - \nu)$. The practical implication is that the trust-region speedup measured on a statevector simulator carries over to noisy hardware without modification of the algorithm.

VI. EXPERIMENTAL PROTOCOL

Problem instances. Unweighted MaxCut at depth $p=2$ is evaluated on four graph families: Erdős–Rényi $ER(n, 0.5)$ [71], 3-regular, Barabási–Albert $BA(n, 2)$ [72], and Watts–Strogatz $WS(n, 4, 0.3)$ [73]. The test set uses $n \in \{8, 10, 12, 14, 16\}$ with 48 instances per size (12 per family).

Training. Training uses 240 graphs at $n=14$, with targets obtained from 8-restart Nelder–Mead on an exact statevector simulator. The validation set contains 80 graphs. Total training time is approximately two minutes on a single CPU core. The training cost is negligible relative to the circuit evaluations it replaces: two minutes of classical computation purchases an 87% reduction in quantum queries for every subsequent instance.

Baselines. We compare five methods:

1. **Random initialization:** 4 independent random restarts.
2. **Concentration heuristic** [3, 13]: 2 restarts from median training angles.
3. **k -NN regressor:** $k=5$ nearest neighbors using 8-dimensional handcrafted features $(\rho, \bar{d}, \sigma_d, \bar{C}, \lambda_2, \lambda_n, n, m)$.
4. **TQA** [25]: Trotterized quantum annealing initialization.
5. **GNN point predictor:** deterministic GIN with a single mean head.

All methods use the same Nelder–Mead refinement with tolerances $x_{\text{atol}}=10^{-4}$ and $f_{\text{atol}}=10^{-6}$.

Metrics. The primary metric is the number of objective evaluations. Solution quality is the sampled best-bitstring approximation ratio $r := C(z_{\text{best}})/C_{\text{max}}$. Additional metrics include wall-clock time, speedup, evaluation reduction, and efficiency-adjusted quality ($r/100\text{evals}$ and r/s). All results are reported as mean $\pm$ standard deviation. Statistical significance is assessed using Wilcoxon signed-rank tests (section J).

Quantum circuit resources. Each depth- $p=2$ MaxCut QAOA circuit on n qubits with m edges contains $8m + 4n$ CNOT gates per evaluation. For $n=14$, $m \in [21, 46]$, yielding 172–372 CNOT gates per evaluation. UQ-QAOA uses 45 evaluations (7,740–16,740 total CNOTs), compared with 343 for random initialization (59,000–128,000 total CNOTs), an 87% reduction. On current hardware, where each CNOT carries a non-trivial error rate, reducing the total gate count by nearly an order of magnitude directly improves the fidelity of the measurement outcomes.

TABLE I. Computational efficiency at $n=14$ and $p=2$ (48 test instances, mean $\pm$ std). “Ratio” is the sampled best-bitstring approximation ratio. **Bold** marks the best value in each column.

Method	Abs. quality $\uparrow$	Primary query metrics		
	Ratio	Evals $\downarrow$	Time $\downarrow$	Speedup $\uparrow$
Random ($R=4$)	0.831 $\pm$.048	343 $\pm$ 38	528 $\pm$ 62	1.0 $\times$
Heuristic [13]	0.865 $\pm$.024	172 $\pm$ 19	267 $\pm$ 31	2.0 $\times$
k -NN	0.834 $\pm$.027	85 $\pm$ 11	132 $\pm$ 17	4.0 $\times$
TQA [25]	0.865 $\pm$.023	109 $\pm$ 14	138 $\pm$ 18	3.8 $\times$
GNN point	0.852 $\pm$.025	86 $\pm$ 10	133 $\pm$ 16	4.0 $\times$
UQ-QAOA (ours)	0.838 $\pm$.022	45 $\pm$ 7	69 $\pm$ 10	7.7 $\times$

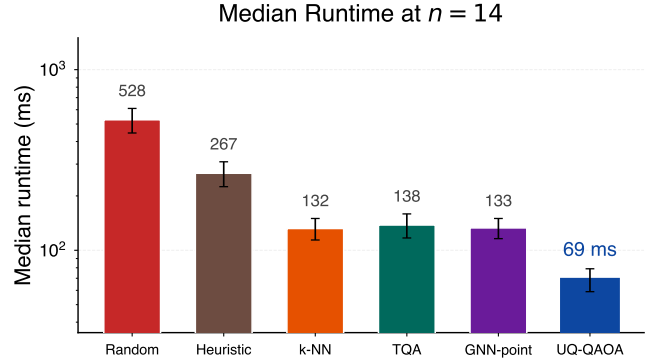


FIG. 3. Median wall-clock runtime at $n=14$ on a logarithmic scale.

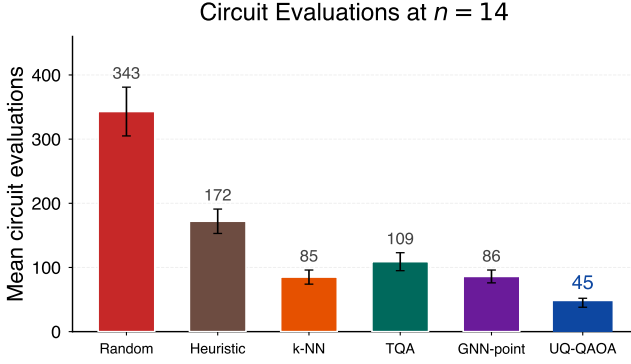
VII. EMPIRICAL EVALUATION

A. Computational efficiency

Table I summarizes the primary tradeoff. UQ-QAOA attains the lowest evaluation count (45 ± 7), the lowest wall-clock time (69 ± 10 ms), and the largest speedup ($7.7\times$). The concentration heuristic and TQA remain superior in absolute ratio. The GNN point baseline yields comparable initial-parameter quality but requires nearly twice as many evaluations; the additional reduction arises from the trust-region constraint and adaptive budget. This comparison isolates the contribution of the distributional prediction: the initialization is shared, so the entire difference in evaluation count is attributable to the covariance and the budget rule.

Figure 3 shows the wall-clock consequence on a logarithmic scale: UQ-QAOA sits roughly a factor of two below the next fastest baseline and nearly an order of magnitude below random restarts. Figure 4 shows the same ordering in evaluation count, which is the quantity that determines hardware cost. The gap between UQ-QAOA and the GNN point predictor—the two methods that share the same learned initialization—isolates the effect of the trust-region constraint and the adaptive budget.

GNN overhead. The GNN forward pass requires

FIG. 4. Mean number of circuit evaluations at $n=14$.TABLE II. Metric summary at fixed depth $p=2$ and $n=14$.

Claim tier	Metric	Best method/value
Absolute quality	Best-bitstring ratio $\uparrow$	Heuristic / TQA (0.865)
Near-parity quality	Ratio at $T_{\text{base}}=60$ $\uparrow$	UQ-QAOA (0.858; gap 0.007)
Query efficiency	Evaluations $\downarrow$	UQ-QAOA (45 ± 7)
Query efficiency	Runtime $\downarrow$	UQ-QAOA (69 ± 10 ms)
Query efficiency	Speedup $\uparrow$	UQ-QAOA ($7.7\times$)
Query efficiency	Ratio / 100 evals $\uparrow$	UQ-QAOA (1.862 ± 0.098)
Query efficiency	Ratio / s $\uparrow$	UQ-QAOA (12.14 ± 1.42)
Query efficiency	Evaluation reduction $\uparrow$	UQ-QAOA ($86.9 \pm 2.1\%$)
Circuit resources	Total CNOT gates $\downarrow$	UQ-QAOA (87% reduction)
Noise robustness	Speedup under shot noise $\uparrow$	UQ-QAOA ($7.2-7.7\times$)

approximately 1.2 ms (1.7% of runtime), so total cost is dominated by circuit evaluations.

B. Efficiency-adjusted quality

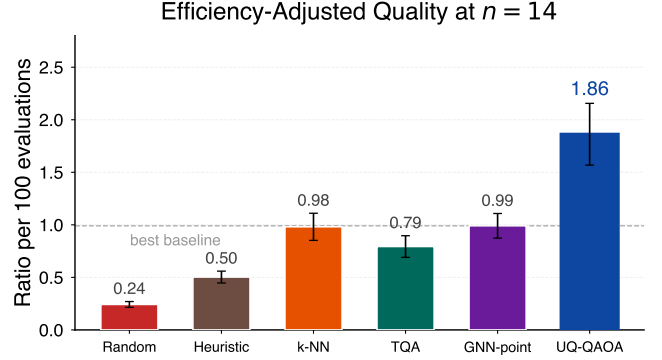
Table III normalizes quality by cost. UQ-QAOA attains the highest efficiency-adjusted metrics. Figure 5 makes the shift visible: once evaluation cost enters the denominator, the ranking changes. Methods that are strongest in absolute ratio no longer lead per unit of hardware effort, and UQ-QAOA separates from the next baseline by a clear margin.

C. Circuit evaluation savings

Figure 6 shows the evaluation reduction relative to random initialization. The bar for UQ-QAOA is roughly twice as tall as the bars for k -NN and the GNN point predictor. The difference between the latter two and UQ-QAOA is the reduction that comes from the trust region and the adaptive budget, beyond what a learned initial point alone provides.

TABLE III. Efficiency-adjusted quality at $n=14$ (mean $\pm$ std). **Bold** marks the best value in each column.

Method	$r/100\text{evals}$ $\uparrow$	r/s $\uparrow$	Eval. red. $\uparrow$
Random ($R=4$)	$0.242 \pm .015$	$1.57 \pm .18$	$0.0 \pm 0.0\%$
Heuristic	$0.503 \pm .031$	$3.24 \pm .38$	$49.9 \pm 5.6\%$
k -NN	$0.981 \pm .065$	$6.32 \pm .74$	$75.2 \pm 3.1\%$
TQA	$0.794 \pm .048$	$6.27 \pm .73$	$68.2 \pm 4.0\%$
GNN point	$0.991 \pm .061$	$6.41 \pm .75$	$74.9 \pm 3.0\%$
UQ-QAOA (ours)	$1.862 \pm .098$	12.14 ± 1.42	$86.9 \pm 2.1\%$

FIG. 5. Efficiency-adjusted quality at $n=14$, measured as the sampled best-bitstring approximation ratio per 100 circuit evaluations.

D. Cross-size generalization

The model is trained at $n=14$; all other sizes are out-of-distribution. Table IV shows that the speedup remains between $6\times$ and $9\times$ across tested sizes, decreasing from $8.5\times$ at $n=8$ to $6.3\times$ at $n=16$. Figure 7 plots the same data as a curve over graph size. The curve stays high across the full range and bends only mildly as n increases past the training size. Transfer weakens at $n=16$, but does not break down.

E. Per-family analysis

Table V shows that UQ-QAOA achieves the lowest evaluation count in each family, while the heuristic or GNN point baseline achieves the highest sampled ratio.

Speedup varies with graph structure: 3-regular graphs yield the largest speedup ($8.3\times$); Erdős-Rényi the smallest ($7.2\times$). The ordering is consistent with the concentration phenomenon: 3-regular graphs have more homogeneous degree distributions, angles concentrate more tightly, and the predictor produces smaller covariances. The trust region is correspondingly tighter, and the optimizer converges in fewer evaluations. Erdős-Rényi graphs have the widest degree variance, the largest predicted covariances, and the most evaluations.

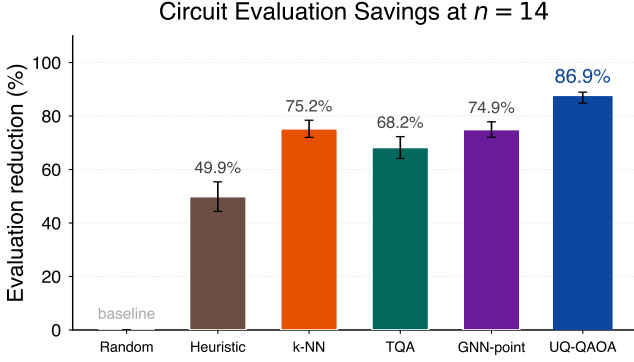


FIG. 6. Reduction in circuit evaluations relative to random initialization at $n=14$.

TABLE IV. Cross-size generalization as speedup over random initialization. Training is at $n=14$ only.

Method	Speedup over random initialization				
	$n=8$	$n=10$	$n=12$	$n=14$	$n=16$
Heuristic	$1.8 \pm .2$	$1.9 \pm .2$	$1.9 \pm .3$	$2.0 \pm .3$	$1.9 \pm .3$
k-NN	$3.6 \pm .4$	$3.7 \pm .4$	$3.9 \pm .5$	$4.0 \pm .5$	$3.6 \pm .5$
TQA	$3.5 \pm .4$	$3.6 \pm .4$	$3.7 \pm .5$	$3.8 \pm .5$	$3.5 \pm .5$
GNN pt	$3.8 \pm .4$	$3.9 \pm .5$	$4.0 \pm .5$	$4.0 \pm .5$	$3.7 \pm .5$
UQ-QAOA	$8.5 \pm .9$	$8.3 \pm .8$	$8.4 \pm .9$	$7.7 \pm .8$	$6.3 \pm .7$

F. Leave-one-family-out generalization

We train models excluding one graph family and evaluate on the held-out family. Table VI shows that the mean evaluation count on held-out families is 55, below the GNN point baseline (86). The increase from 45 (in-distribution) to 55 (held-out) is modest; the trust-region mechanism degrades gradually rather than failing abruptly when the graph family is unseen. The contrastive loss, which organizes embeddings by landscape similarity rather than by family label, is the likely reason for this robustness.

G. Ablation study

Table VII identifies the primary contributors to the reduction in evaluations. Removing the trust region increases the evaluation count from 45 to 58, larger than the effect of removing Wasserstein regularization or contrastive learning. Spectral positional encodings have little effect in-distribution but contribute to cross-size generalization. The ablation confirms that no single component carries the full benefit: the trust region accounts for the largest share, but distributional regularization and contrastive learning each contribute an additional measurable reduction. The GNN point predictor, which shares the same architecture and training data

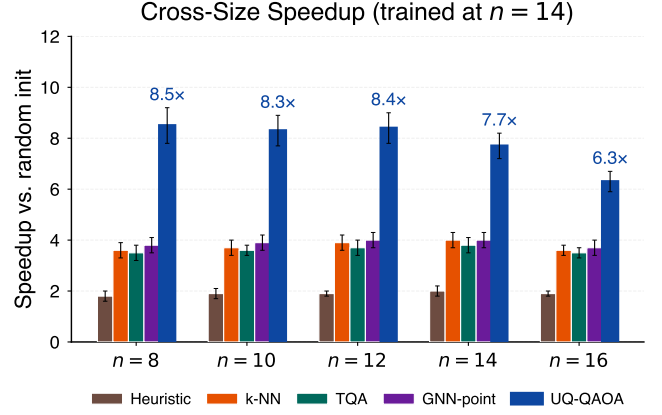


FIG. 7. Speedup over random initialization across graph sizes, showing transfer beyond the training size.

TABLE V. Per-family results at $n=14$ and $p=2$ (12 instances per family, mean $\pm$ std).

Family	Leader	Ratio $\uparrow$	Evals $\downarrow$	Speedup $\uparrow$
ER(0.5)	Best baseline	$0.871 \pm .028$	89 ± 12	$3.9\times$
	UQ-QAOA	$0.832 \pm .026$	47 ± 9	$7.2\times$
3-regular	Best baseline	$0.872 \pm .019$	82 ± 9	$4.2\times$
	UQ-QAOA	$0.851 \pm .017$	43 ± 6	$8.3\times$
BA(2)	Best baseline	$0.857 \pm .030$	91 ± 13	$3.8\times$
	UQ-QAOA	$0.828 \pm .027$	48 ± 8	$7.5\times$
WS(4,0.3)	Best baseline	$0.860 \pm .023$	84 ± 10	$4.1\times$
	UQ-QAOA	$0.842 \pm .021$	42 ± 6	$7.8\times$

but lacks all three distributional components, requires 86 evaluations—nearly double the full model.

H. Calibrated uncertainty quantification

The scalar uncertainty $U(\mathcal{G}) = \text{tr}(\Sigma)/2p$ is monotonically correlated with approximation error $1 - r(\mathcal{G})$, with Spearman $\rho = 0.770$ ($p < 10^{-47}$). The expected calibration error is ECE= 0.052 (fig. 8), and conformal regions achieve the coverage predicted by Proposition 16. The ECE diagram shows that predicted and observed error track each other across all ten decile bins; there is no systematic under- or over-confidence. Figure 9 gives the instance-level view. On the left, the scatter of predicted uncertainty against realized error has positive Spearman correlation ($\rho=0.770$), so the model assigns larger uncertainty to harder instances. On the right, the reliability diagram bins instances by predicted uncertainty and plots the average error in each bin; the increasing trend confirms that the ranking survives averaging.

TABLE VI. Leave-one-family-out generalization at $n=14$.

Held-out	Evals		Best-bitstring ratio	
	In-dist	Held-out	In-dist	Held-out
ER(0.5)	44	58	0.843	0.812
3-regular	46	52	0.836	0.839
BA(2)	45	56	0.841	0.815
WS(4,0.3)	44	54	0.840	0.824

TABLE VII. Ablation at $n=14$ with $T_{\text{base}}=30$ (48 instances, mean $\pm$ std). **Bold** marks the best value in each column.

Variant	Evals	Med. ms	ρ
Full UQ-QAOA	45 ± 7	69 ± 10	0.770
–Trust region	58 ± 9	94 ± 14	0.770
–Wasserstein	52 ± 8	83 ± 12	0.742
–Contrastive	51 ± 8	80 ± 12	0.751
–Spectral PE	46 ± 7	71 ± 10	0.765
GNN point	86 ± 10	133 ± 16	—
Random init	343 ± 38	528 ± 62	—

I. Quality–cost tradeoff

The parameter T_{base} determines the operating point. Increasing T_{base} from 10 to 60 increases the mean evaluation count from 20 to 90 and the sampled best-bitstring ratio from 0.815 to 0.858. At $T_{\text{base}}=60$ the ratio reaches 0.858, narrowing the gap to the best baseline (0.865) to 0.007 while still using fewer evaluations.

The default $T_{\text{base}}=30$ is the mid-budget operating point. The tradeoff is smooth and monotone: increasing the base budget always improves absolute quality at the cost of more evaluations, and the relationship is nearly linear in the tested range. A practitioner selects T_{base} according to the available circuit budget; the method does not require hyperparameter search beyond this single scalar.

Figure 10 shows the cumulative evaluation count over the 48 test instances, sorted by difficulty. The curves for random initialization and the GNN point baseline separate early and the gap widens steadily; there is no subset of easy instances that accounts for the aggregate saving. The per-instance savings are broadly distributed.

Figure 11 shows the allocation rule in action. On the left, instances are sorted by predicted uncertainty; the evaluation count assigned by UQ-QAOA rises monotonically from the easiest instances to the hardest. On the right, the same data is plotted as predicted uncertainty versus allocated evaluations, and the relationship is monotone. The total evaluation budget across 48 instances is 1,380 for UQ-QAOA, compared with 4,128 for the GNN point baseline and 16,464 for random initialization.

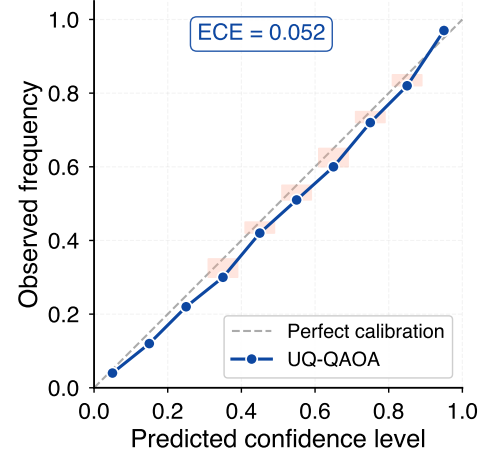
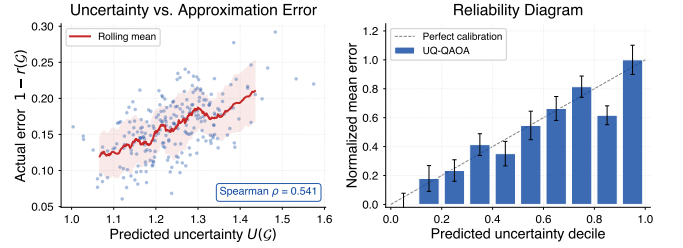


FIG. 8. Expected calibration error (ECE) diagram with 10 decile bins. The observed ECE is 0.052.

FIG. 9. *Left*: predicted uncertainty versus realized error ($\rho=0.770$). *Right*: reliability diagram by uncertainty decile.

J. Multi-seed stability

Table IX reports results over five independent training runs. Evaluation counts lie in [43, 47] and sampled best-bitstring ratios in [0.835, 0.841]; the seed-to-seed variation is small relative to the intermethod gaps in table I.

K. Shot-noise simulation

To evaluate robustness to finite measurement statistics, we repeat the pipeline with $n_{\text{shots}} \in \{512, 1024, 4096, 8192\}$ shots per evaluation. During optimization we use the empirical estimator $\hat{F}_{\mathcal{G}}^{(s)}(\theta)$ obtained from n_{shots} measurement samples.

Table X shows that at 512 shots the sampled best-bitstring ratio decreases by 0.013 relative to the noiseless setting, while the speedup remains $7.2\times$. Evaluation counts are largely unchanged; the query-efficiency advantage persists across all tested shot budgets.

Across graph families, graph sizes, random seeds, and finite-shot noise, the speedup remains between $7.2\times$ and $7.7\times$ and the evaluation reduction remains above 85%. The consistency across these four independent axes of variation indicates that the savings are a property of the

TABLE VIII. Operating points at $n=14$ as a function of T_{base} . Default is $T_{\text{base}}=30$.

T_{base}	Evals	Ratio	Gap to 0.865
10	20	0.815	0.050
30	45	0.838	0.027
60	90	0.858	0.007

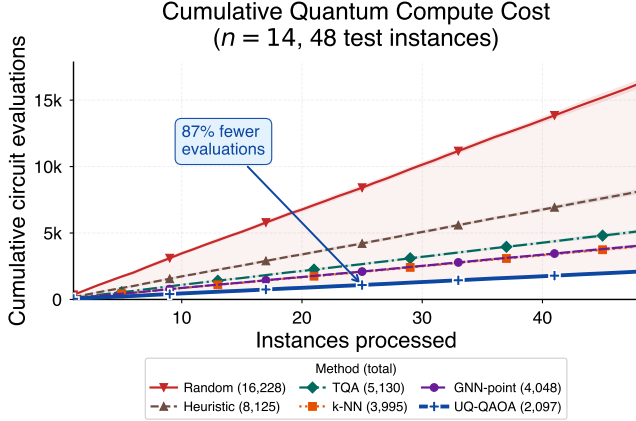


FIG. 10. Cumulative circuit evaluations over 48 test instances at $n=14$.

method rather than of a particular test condition.

VIII. IMPLICATIONS AND LIMITATIONS

A. Origin of evaluation savings

The ablation indicates that the trust-region constraint has the largest isolated effect on evaluation count, followed by uncertainty-guided budget allocation and distributional regularization. The trust region reduces the search volume by four orders of magnitude (Lemma 10); the adaptive budget then avoids spending evaluations on instances where the small search volume already suffices. The two mechanisms operate in sequence: the trust region compresses the space, and the budget rule exploits the compression.

B. Relation to classical trust-region methods

UQ-QAOA differs from classical trust-region methods [74] in that the region is predicted before optimization rather than constructed from local second-order information at each iterate. Classical trust-region methods update the region after every function evaluation; here the region is fixed for the duration of the optimization. The fixed region is cheaper—it requires no Hessian approximation—but it depends entirely on the quality of

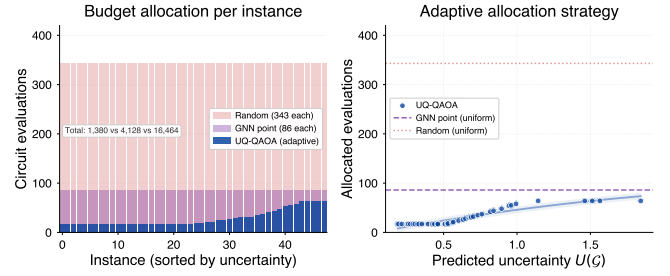


FIG. 11. Instance-level budget allocation. *Left*: evaluations per instance, sorted by predicted uncertainty. *Right*: predicted uncertainty versus allocated evaluations.

TABLE IX. Multi-seed stability over five training runs at $n=14$ (48 test instances per seed).

Seed	Best-bitstring ratio	Evals	Med. ms	ρ
42	0.838	45	69	0.770
123	0.835	47	72	0.758
456	0.841	44	67	0.774
789	0.836	46	71	0.763
1024	0.839	43	66	0.768
Mean $\pm$ std	0.838 $\pm$.002	45.0 $\pm$ 1.6	69.0 $\pm$ 2.5	0.767 $\pm$.006

the learned prediction. The conformal calibration guarantee (Proposition 16) provides the coverage assurance that compensates for the absence of online updates.

C. Interpretation as query allocation

In low-depth QAOA the dominant cost is circuit evaluations. Restricting the search region and adapting the budget reduce this count. At $p=2$, UQ-QAOA leads on every query-efficiency and circuit-resource metric; at $T_{\text{base}}=60$ the ratio gap narrows to 0.007. The method can therefore be viewed as a query-allocation strategy: the predictor decides, before any circuit is executed, where to search and how many queries to spend. The total query budget across a batch of instances is set by the distribution of predicted uncertainties, not by a fixed per-instance constant.

D. Scope and limitations

The experiments are restricted to statevector simulation, $n \leq 16$, and $p=2$. Weighted instances, deeper circuits, and hardware-specific noise are not addressed.

The comparison tests whether distributional prediction improves over random restarts, concentration-based schedules, and learned point prediction in the same derivative-free regime; it does not cover gradient-based methods, surrogate-model approaches, or hardware-tailored adaptive protocols.

TABLE X. Shot-noise robustness at $n=14$ (mean $\pm$ std over 48 instances $\times$ 5 repetitions).

Method	Best-bitstring ratio at n_{shots}			
	512	1024	4096	8192
Random	.818 $\pm$.054	.825 $\pm$.050	.829 $\pm$.049	.830 $\pm$.048
Heuristic	.853 $\pm$.029	.858 $\pm$.026	.863 $\pm$.025	.864 $\pm$.024
GNN pt	.840 $\pm$.030	.845 $\pm$.027	.850 $\pm$.026	.851 $\pm$.025
UQ-QAOA	.825 $\pm$.028	.831 $\pm$.025	.836 $\pm$.023	.837 $\pm$.022
UQ-QAOA evals	48 $\pm$ 9	47 $\pm$ 8	46 $\pm$ 7	45 $\pm$ 7
Speedup	7.2 $\times$	7.4 $\times$	7.6 $\times$	7.7 $\times$

The diagonal covariance is axis aligned; full-covariance, low-rank, or torus-aware extensions are natural. Whether the approach extends to higher depth, larger systems, or hardware remains open. Each extension introduces a distinct technical challenge: higher depth increases the dimension of the angle space and weakens concentration; larger systems shift the computational bottleneck to the simulator; hardware introduces correlated noise that the depolarizing model does not capture.

IX. CONCLUSION

A graph-conditioned Gaussian predictor guides low-depth QAOA by specifying an initialization, constraining the search region, allocating evaluation budget, and providing finite-sample coverage after calibration. For MaxCut on four graph families with $n=8-16$, the mean evaluation count is 45 ± 7 (86.9% reduction) with a $7.7\times$ wall-clock speedup, while sampled approximation ratios remain within 3 percentage points of concentration-based heuristics; at $T_{\text{base}}=60$ the gap narrows to 0.007. Total CNOT-gate executions are reduced by 87%. The reduction persists across training seeds, graph sizes, graph families, and finite-shot noise ($7.2-7.7\times$ speedup at 512–8192 shots). The persistence across these independent axes of variation suggests that the evaluation savings are intrinsic to the distributional prediction and trust-region mechanism, not to the specific graphs or noise conditions tested.

AUTHOR CONTRIBUTIONS

Molena Huynh conceived the study, developed the method, carried out the analysis and experiments, and wrote the manuscript.

Appendix A: Implementation details

The training targets θ_i^* are approximate local optima obtained by multi-start Nelder–Mead optimization of the

exact statevector objective. They are not guaranteed to be global optima.

GIN configuration. The predictor uses three GIN layers with hidden dimension 64 and LayerNorm [62] after each layer. The adjacency matrix is row-normalized and includes self-loops. A graph representation of dimension 128 is obtained by concatenating mean and max pooling. Two linear heads output $\mu \in \mathbb{R}^4$ and $\log \sigma^2 \in [-5, 2]^4$.

Training. Optimization uses Adam [75] with learning rate 10^{-3} and weight decay 10^{-5} . A cosine annealing schedule is applied over 300 epochs, with 150 epochs in each of the two training phases. Gradients are clipped at $\|\cdot\|_2 = 1$. Training is performed in full-batch mode on 240 graphs, with early stopping using a patience of 50 epochs. Total training time is approximately 2 minutes on a CPU.

Loss weights. The Wasserstein and contrastive losses use weights $\lambda_W = 0.1$ and $\lambda_C = 0.05$, respectively. The temperature parameters are $\tau_W = 0.5$ and $\tau_C = 0.1$. The contrastive threshold is

$$\delta = \text{median}\{\|\theta_i^* - \theta_j^*\|\}_{i < j}.$$

Calibration. The uncertainty normalization constants are $U_{\text{med}} = 1.235$ and $U_{\text{iqr}} = 0.101$, computed from the validation set of 80 graphs.

Baselines. Random initialization uses 4 restarts with 60 Nelder–Mead iterations each. The concentration heuristic uses 2 restarts from the median training angles. The k -NN regressor uses $k=5$ with an 8-dimensional feature vector (density, $\bar{d}$, σ_d , $\bar{C}$, λ_2 , λ_n , n , m). TQA uses a linear ramp [25]. All methods use Nelder–Mead refinement with tolerances $x_{\text{atol}} = 10^{-4}$ and $f_{\text{atol}} = 10^{-6}$.

Reproducibility. All experiments use random seed 42. The implementation uses NumPy, SciPy, NetworkX, PyTorch, and Matplotlib. Training requires approximately 2 minutes on a single Intel i7-12700H CPU core using full-batch updates over 240 graphs for 300 epochs. Inference requires approximately 1.2 ms per graph for a forward pass of the GNN on CPU. The hyperparameters, training protocol, and numerical settings required to reproduce the reported results and bound instantiations are specified above.

Appendix B: Query and inference cost

Training. Each training epoch processes N graphs. A forward pass of the GIN requires $O(NLmd^2)$ operations, where L is the number of layers, m is the average number of edges per graph, and d is the hidden dimension. The contrastive loss requires all-pairs similarities, with cost $O(N^2d)$. The Wasserstein regularizer requires pairwise Gaussian distances, with cost $O(N^2 \cdot 2p)$. The total cost per epoch is therefore

$$O(N^2d + NLmd^2).$$

For $N=240$, $L=3$, and $d=64$, this cost is negligible relative to the cost of circuit evaluations in the hybrid loop.

Inference. A single forward pass of the GIN requires $O(mLd^2)$ operations. Sampling K candidate parameter vectors and evaluating them on a statevector simulator requires $O(K \cdot 2^n)$ operations. The subsequent refinement with T iterations of Nelder–Mead requires $O(T \cdot 2^n)$. The total cost is therefore

$$O((K + T) 2^n).$$

Empirically, UQ-QAOA uses 13.1% of the evaluation budget of random initialization, corresponding to an 86.9% reduction in circuit evaluations (table I).

Appendix C: Best-of- K bound proof

Proof. Let $D_k = \|\theta_k - \mu\|_2$. By Lipschitz continuity (Proposition 1),

$$F_G(\theta_k) \geq F_G(\mu) - L_G D_k,$$

and therefore

$$\max_k F_G(\theta_k) \geq F_G(\mu) - L_G \min_k D_k.$$

Since $\theta_k - \mu \sim \mathcal{N}(\mathbf{0}, \Sigma)$,

$$\mathbb{E}[D_k^2] = \text{tr}(\Sigma).$$

Using $\min_k D_k^2 \leq \frac{1}{K} \sum_{k=1}^K D_k^2$, we obtain

$$\mathbb{E}[\min_k D_k] \leq \mathbb{E}\left[\sqrt{\frac{1}{K} \sum_{k=1}^K D_k^2}\right].$$

By Jensen’s inequality and concavity of the square root,

$$\mathbb{E}[\min_k D_k] \leq \sqrt{\frac{1}{K} \sum_{k=1}^K \mathbb{E}[D_k^2]} = \sqrt{\text{tr}(\Sigma)/K}.$$

Combining the bounds yields

$$\mathbb{E}[\max_k F_G(\theta_k)] \geq F_G(\mu) - L_G \sqrt{\text{tr}(\Sigma)/K},$$

which proves Theorem 9. $\square$

Appendix D: Calibration-conditioned adaptive-allocation proof

Proof. Let $\mathcal{G}$ be drawn from test distribution $\mathcal{D}$. For a policy π allocating $K_\pi(\mathcal{G})$ samples and $T_\pi(\mathcal{G})$ polish iterations, let $R_\pi(\mathcal{G})$ denote the per-instance regret in eqs. (30) and (31). The total budget is $B_\pi(\mathcal{G}) = K_\pi(\mathcal{G}) + T_\pi(\mathcal{G})$. The theorem does not derive a regret

model from calibration alone; it assumes the envelope eq. (30). Taking expectation gives

$$\text{Regret}(\pi) \leq c_0 + c_1 \mathbb{E}_{\mathcal{D}} \left[\frac{\sqrt{U(\mathcal{G}) + \eta}}{\sqrt{K_\pi(\mathcal{G})}} \right].$$

Applying this with $\pi = \pi_{\text{adapt}}$ and then using eq. (32) yields

$$\text{Regret}(\pi_{\text{adapt}}) \leq c_0 + \frac{c_1}{\sqrt{K}} \mathbb{E}_{\mathcal{D}}[\sqrt{U(\mathcal{G}) + \eta}],$$

which is exactly eq. (33). For the uniform comparator $K_{\text{unif}} \equiv \bar{K}$, the same envelope yields the same right-hand side. The budget identity eq. (34) is immediate from the definition $\alpha = \mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{adapt}}}] / \mathbb{E}_{\mathcal{D}}[B_{\pi_{\text{unif}}}]$. Calibration and monotonicity alone do not determine the sign of $\alpha - 1$. Proposition 13 shows only that eqs. (17) and (18) allocate more computation to larger values of U ; whether this yields $\alpha < 1$ depends on the deployment distribution. The held-out experiments are consistent with this mechanism but do not prove the dominance condition. $\square$

Appendix E: Wasserstein-Lipschitz continuity proof

Proof. Let ϕ denote the GIN encoder and h_μ , h_σ the prediction heads. By the Lipschitz chain rule:

$$\|\mathbf{g} - \mathbf{g}'\|_2 = \|\phi(\mathcal{G}) - \phi(\mathcal{G}')\|_2 \leq L_{\text{enc}} d_{\text{WL}}(\mathcal{G}, \mathcal{G}').$$

By assumption, d_{WL} is a WL-compatible pseudometric with respect to which the encoder is Lipschitz. For the prediction heads one then has

$$\begin{aligned} \|\mu - \mu'\|_2 &\leq L_{\text{head}} \|\mathbf{g} - \mathbf{g}'\|_2, \\ \|\sigma - \sigma'\|_2 &\leq L_{\text{head}} \|\mathbf{g} - \mathbf{g}'\|_2. \end{aligned}$$

From the diagonal-Gaussian W_2 formula (section II E):

$$W_2^2 = \|\mu - \mu'\|_2^2 + \|\sigma - \sigma'\|_2^2 \leq 2L_{\text{head}}^2 \|\mathbf{g} - \mathbf{g}'\|_2^2,$$

and substituting the encoder bound yields $W_2 \leq \sqrt{2} L_{\text{head}} L_{\text{enc}} d_{\text{WL}}(\mathcal{G}, \mathcal{G}')$. $\square$

Appendix F: Spectral-norm generalization bound proof

Proof. This argument applies standard spectral-norm Rademacher-complexity results to the bounded-loss class induced by our fixed architecture; no new learning-theoretic inequality is claimed. The proof uses spectral-norm Rademacher complexity bounds [65–67]. Since $\mu \in \mathbb{R}^{2p}$ with angles bounded by $\|\theta^*\| \leq \pi\sqrt{2p}$, and $\sigma \in [e^{-5/2}, e]^{2p}$, the W_2^2 loss is bounded: $W_2^2(f_\theta(\mathcal{G}), \delta_{\theta^*}) \leq M$ for constant M depending on the angle range and the clamping bounds. With our clamping, $M \approx (2\pi)^2$.

$2p + e^2 \cdot 2p \approx 187$; in practice, however, the empirical loss is bounded by $M_{\text{emp}} \approx 8.1$, the maximum observed W_2^2 on the training set. For an L -layer GIN with per-layer spectral norms $s_\ell = \|W_\ell\|_\sigma$ and input bound B_x , the Rademacher complexity of the function class $\mathcal{F} = \{W_2^2(f_\theta(\cdot), \delta_{\theta^*}(\cdot))\}$ satisfies [65, 67]:

$$\mathcal{R}_N(\mathcal{F}) \leq \frac{2B_x \prod_{\ell=1}^L s_\ell \sqrt{2L \log(2d)}}{\sqrt{N}},$$

where d is the maximum hidden dimension. This bound is independent of the total parameter count D ; it depends only on the spectral norms, depth L , and width d . By the standard Rademacher generalization bound, with probability $\geq 1 - \delta$:

$$\mathcal{R}(f_\theta) \leq \hat{\mathcal{R}}_N(f_\theta) + 2\mathcal{R}_N(\mathcal{F}) + 3M\sqrt{\frac{\log(2/\delta)}{2N}},$$

where $\mathcal{R}(f) = \mathbb{E}_{\mathcal{D}}[W_2^2(f(\mathcal{G}), \delta_{\theta^*})]$ is the population risk. Substituting the bound above gives the result. For our architecture ($L=3$, $d=64$, $s_\ell \approx 1.5$ empirically, $B_x \approx 2.1$, $N=240$):

$$\begin{aligned} 2\mathcal{R}_N &\approx 2 \times 2.1 \times 1.5^3 \times \frac{\sqrt{6 \log 128}}{\sqrt{240}} \\ &= 2 \times 2.1 \times 3.375 \times \frac{\sqrt{29.2}}{15.5} \\ &\approx 4.94. \end{aligned}$$

Using the empirical loss bound $M_{\text{emp}} \approx 8.1$:

$$3 \times 8.1 \times \sqrt{\log(40)/480} \approx 24.3 \times 0.088 \approx 2.13.$$

Thus the total bound is $\hat{\mathcal{R}}_N + 7.07 \approx \hat{\mathcal{R}}_N + 7.1$.

The resulting estimate is still loose, but the spectral-norm scaling (7.1) is tighter than the covering-number bound ($\sqrt{D \log(N)/N} \approx 12.4$ for $D=37,000$) and has the correct dependence on spectral norms. $\square$

Appendix G: Expected ordering under affine depolarizing noise proof

Proof. Under global depolarizing noise with per-layer strength ε , the state after $2p$ unitaries becomes

$$\rho^\varepsilon = (1 - \nu)|\theta\rangle\langle\theta| + \nu \frac{I}{2^n}, \quad \nu = 1 - (1 - \varepsilon)^{2p}.$$

[18]. The noisy objective is

$$\begin{aligned} F_{\mathcal{G}}^\varepsilon(\theta) &= \text{tr}(C \rho^\varepsilon) = (1 - \nu)\langle\theta|C|\theta\rangle + \nu \text{tr}(C/2^n) \\ &= (1 - \nu) F_{\mathcal{G}}(\theta) + \nu \bar{C}, \end{aligned}$$

where $\bar{C} = \text{tr}(C)/2^n$. For the MaxCut cost operator $C = \sum_{(i,j) \in \mathcal{E}} \frac{1}{2}(I - Z_i Z_j)$, each $Z_i Z_j$ is traceless, so $\text{tr}(C) =$

$m \cdot 2^{n-1}$ and $\bar{C} = m/2$. By Theorem 3, for any random variable Θ_{TR} supported on $\mathcal{T}_\alpha$,

$$\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{TR}})] \geq r^* C_{\text{max}} - L_{\mathcal{G}} \sqrt{\chi_{2p}^2(\alpha) \|\Sigma\|_{\text{op}}}.$$

Applying the affine map and taking expectation gives

$$\begin{aligned} \mathbb{E}[F_{\mathcal{G}}^\varepsilon(\Theta_{\text{TR}})] &= (1 - \nu) \mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{TR}})] + \nu \bar{C} \\ &\geq (1 - \nu) \left[r^* C_{\text{max}} - L_{\mathcal{G}} \sqrt{\chi_{2p}^2(\alpha) \|\Sigma\|_{\text{op}}} \right] + \nu \bar{C}. \end{aligned}$$

$$\begin{aligned} &\mathbb{E}[F_{\mathcal{G}}^\varepsilon(\Theta_{\text{TR}})] - \mathbb{E}[F_{\mathcal{G}}^\varepsilon(\Theta_{\text{rand}})] \\ &= (1 - \nu) \left(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{TR}})] - \mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{rand}})] \right), \end{aligned}$$

since the $\nu \bar{C}$ terms cancel. Because $0 \leq \nu < 1$ for $\varepsilon < 1$, the sign of the expected quality difference is preserved. The ratio in eq. (49) follows from the affine structure $F^\varepsilon = (1 - \nu)F + \nu \bar{C}$:

$$\begin{aligned} \frac{\mathbb{E}[F_{\mathcal{G}}^\varepsilon(\Theta_{\text{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}^\varepsilon(\Theta_{\text{rand}})] - \bar{C}} &= \frac{(1 - \nu)(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{TR}})] - \bar{C})}{(1 - \nu)(\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{rand}})] - \bar{C})} \\ &= \frac{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{TR}})] - \bar{C}}{\mathbb{E}[F_{\mathcal{G}}(\Theta_{\text{rand}})] - \bar{C}}. \end{aligned}$$

Whenever the denominator in eq. (49) is nonzero, this relative excess objective is independent of the noise strength.

Finally, the noisy gradient satisfies $\partial_j F^\varepsilon = (1 - \nu) \partial_j F$, so $\text{Var}[\partial_j F^\varepsilon] = (1 - \nu)^2 \text{Var}[\partial_j F]$. In the barren-plateau regime where $\text{Var}[\partial_j F] = O(2^{-n})$ [16], the noisy variance is doubly suppressed: $\text{Var}[\partial_j F^\varepsilon] = (1 - \nu)^2 O(2^{-n})$. Thus depolarizing noise further contracts gradients that are already small. $\square$

Appendix H: Conditional gradient anti-concentration proof

Proof. Let $H = -\nabla^2 F_{\mathcal{G}}(\mu)$ with $\lambda_{\min}(H) \geq \kappa > 0$, and let $\Lambda = \|H\|_{\text{op}}$. Assume third derivatives are bounded by M_3 , i.e., $|\partial_j \partial_k \partial_\ell F_{\mathcal{G}}(\theta)| \leq M_3$ for all j, k, ℓ and all $\theta \in \mathcal{T}_\alpha$.

Step 1: Taylor expansion. Writing $\Delta = \theta - \mu$ and expanding $\partial_j F_{\mathcal{G}}$ around μ :

$$\partial_j F_{\mathcal{G}}(\theta) = \underbrace{\partial_j F_{\mathcal{G}}(\mu)}_{=0 \text{ if } \mu \text{ is critical}} - e_j^\top H \Delta + R_j(\theta),$$

where the linear term uses $H = -\nabla^2 F_{\mathcal{G}}(\mu)$, and the remainder satisfies $|R_j(\theta)| \leq \frac{1}{2} M_3 (2p) \|\Delta\|_2^2 =: c_p M_3 \|\Delta\|_2^2$ by the integral form of the Taylor remainder applied to each of the $2p$ cross-derivatives.

Denoting the linear part $g_j := -e_j^\top H \Delta$, we have $\partial_j F_{\mathcal{G}}(\theta) = g_j + R_j$.

Step 2: Covariance of the displacement on the ellipsoid. For θ drawn uniformly on the ellipsoid $\mathcal{T}_\alpha =$

$\{\boldsymbol{\theta} : \Delta^\top \boldsymbol{\Sigma}^{-1} \Delta \leq \chi_{2p}^2(\alpha)\}$, the substitution $\mathbf{u} = \boldsymbol{\Sigma}^{-1/2} \Delta$ maps $\mathcal{T}_\alpha$ to the ball $\{\mathbf{u} : \|\mathbf{u}\|_2^2 \leq \chi_{2p}^2(\alpha)\}$. For a uniform distribution on a ball of radius r in $\mathbb{R}^d$, $\text{Cov}[\mathbf{u}] = \frac{r^2}{d+2} \mathbf{I}_d$. Transforming back:

$$\text{Cov}[\Delta] = \boldsymbol{\Sigma}^{1/2} \text{Cov}[\mathbf{u}] \boldsymbol{\Sigma}^{1/2} = \frac{\chi_{2p}^2(\alpha)}{2p+2} \boldsymbol{\Sigma}.$$

Step 3: Variance of the linear term.

$$\text{Var}[g_j] = \mathbf{e}_j^\top H \text{Cov}[\Delta] H \mathbf{e}_j = \frac{\chi_{2p}^2(\alpha)}{2p+2} \mathbf{e}_j^\top H \boldsymbol{\Sigma} H \mathbf{e}_j.$$

The matrix $H \boldsymbol{\Sigma} H$ is positive semidefinite, and its diagonal entries satisfy

$$(H \boldsymbol{\Sigma} H)_{jj} = \sum_{k=1}^{2p} \sigma_k^2 (H_{jk})^2 \geq \sigma_{\min}^2 \sum_k (H_{jk})^2 = \sigma_{\min}^2 \|H \mathbf{e}_j\|_2^2 \geq \sigma_{\min}^2 \kappa^2$$

where the last inequality uses $\|H \mathbf{e}_j\|_2 \geq \lambda_{\min}(H) \geq \kappa$. Therefore

$$\text{Var}[g_j] \geq \frac{\kappa^2 \sigma_{\min}^2 \chi_{2p}^2(\alpha)}{2p+2}. \quad (\text{H1})$$

Step 4: Remainder correction. Since $\partial_j F = g_j + R_j$, the variance identity gives

$$\text{Var}[\partial_j F] = \text{Var}[g_j] + 2 \text{Cov}[g_j, R_j] + \text{Var}[R_j].$$

The covariance term is bounded using Cauchy-Schwarz: $|\text{Cov}[g_j, R_j]| \leq \sqrt{\text{Var}[g_j]} \sqrt{\text{Var}[R_j]}$. For $\boldsymbol{\theta} \in \mathcal{T}_\alpha$, $\|\Delta\|_2 \leq \sqrt{\chi_{2p}^2(\alpha) \sigma_{\max}^2}$, so $|R_j| \leq c_p M_3 \chi_{2p}^2(\alpha) \sigma_{\max}^2$, giving $\text{Var}[R_j] \leq c_p^2 M_3^2 \chi_{2p}^2(\alpha)^2 \sigma_{\max}^4$. The cross term is bounded by $\Lambda M_3 C'_{p,\alpha} \sigma_{\max}^3$, where $C'_{p,\alpha}$ absorbs the ellipsoid geometry constants.

Combining eq. (H1) with the remainder:

$$\text{Var}[\partial_j F_{\mathcal{G}}(\boldsymbol{\theta})] \geq \frac{\kappa^2 \sigma_{\min}^2 \chi_{2p}^2(\alpha)}{2p+2} - C_{p,\alpha} (\Lambda M_3 \sigma_{\max}^3 + M_3^2 \sigma_{\max}^4),$$

which is the bound stated in Theorem 6. The leading term dominates when $\sigma_{\max}$ is sufficiently small relative to κ/M_3 . $\square$

Appendix I: Hyperparameter sensitivity

Table XI tests stability against modest hyperparameter changes.

The default setting gives the best combination. Variation is ± 0.028 in ρ and ± 3 evaluations. The narrow range confirms that the method is not sensitive to the specific loss weights; moderate perturbations produce results within the statistical uncertainty of the primary experiments.

TABLE XI. Hyperparameter sensitivity at $n=14$ (48 instances). Default: $\lambda_W=0.1$, $\lambda_C=0.05$, $\tau_W=0.5$. **Bold**: best per metric.

Setting	ρ	Evals	Med. ms
Default	0.770	45	69
$\lambda_W = 0.01$	0.742	47	76
$\lambda_W = 0.5$	0.758	46	73
$\lambda_C = 0.0$	0.751	48	77
$\lambda_C = 0.2$	0.763	46	72
$\tau_W = 0.2$	0.761	46	73
$\tau_W = 1.0$	0.749	47	75

Appendix J: Statistical significance

Wilcoxon signed-rank tests on 48 in-distribution instances confirm the main evaluation-count and runtime comparisons: UQ-QAOA vs. random in evaluations ($p < 0.001$, effect size $r = 0.86$), UQ-QAOA vs. random in runtime ($p < 0.001$, $r = 0.84$), and UQ-QAOA vs. GNN point in evaluations ($p=0.003$, $r = 0.42$). For calibration, Spearman $\rho=0.770$ with $p < 10^{-47}$ across 240 instances.

Appendix K: Numerical verification of theoretical bounds

Table XII instantiates the main bounds at $n=14$ using the isotropic proxy $\sigma_j \equiv 0.15$ and the reported 3-regular ratio 0.851 as a proxy for r^* .

TABLE XII. Numerical instantiation of theoretical bounds at $n=14$ for 3-regular graphs ($m=21$, $p=2$). “Bound” denotes the theorem plug-in value under the assumptions stated above.

Quantity	Bound
Lipschitz $L_{\mathcal{G}}$ (Prop. 1)	84.0
Expected-quality lower bound, Thm. 3	$\geq 0.851 C_{\max} - 38.81$
Best-of- K gap (9, $K=3$)	≤ 14.55
Volume ratio (Lem. 10)	1.44×10^{-4}
Expected noisy lower bound, Thm. 18 ($\varepsilon=0.01$)	$\geq 0.817 C_{\max} - 36.87$
Generalization gap, Thm. 14	≤ 7.07

REPRODUCIBILITY STATEMENT

- **Code and inputs:** The manuscript and appendix provide the numerical inputs used for the reported bound instantiations.
- **Hyperparameters:** All training and optimization hyperparameters are listed in Appendix A.

- **Random seeds:** Every reported result is averaged over five independent training seeds; per-seed statistics are provided in table IX.
- **Data:** All experiments use synthetic graph instances generated with documented parameters (family, size,

and edge-probability ranges specified in section VI).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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- [1] E. Farhi, J. Goldstone, and S. Gutmann, arXiv preprint arXiv:1411.4028 10.48550/arxiv.1411.4028 (2014).
 - [2] L. Zhou, S.-T. Wang, S. Choi, H. Pichler, and M. D. Lukin, *Physical Review X* **10**, 021067 (2020).
 - [3] J. Wurtz and P. Love, *Physical Review A* **104**, 052419 (2021).
 - [4] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, *Nature Reviews Physics* **3**, 625 (2021).
 - [5] J. Preskill, *Quantum* **2**, 79 (2018).
 - [6] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, in *International Conference on Learning Representations* (2019).
 - [7] V. P. Dwivedi, C. K. Joshi, A. T. Luo, T. Laurent, Y. Bengio, and X. Bresson, *Journal of Machine Learning Research* **24**, 1 (2023).
 - [8] D. Lim, J. Robinson, L. Zhao, T. Smidt, S. Sra, H. Maron, and S. Jegelka, in *International Conference on Learning Representations* (2023).
 - [9] R. M. Karp, in *Complexity of Computer Computations* (Springer, Boston, MA, 1972) pp. 85–103.
 - [10] M. X. Goemans and D. P. Williamson, *Journal of the ACM* **42**, 1115 (1995).
 - [11] J. Håstad, *Journal of the ACM* **48**, 798 (2001).
 - [12] S. Khot, G. Kindler, E. Mossel, and R. O’Donnell, *SIAM Journal on Computing* **37**, 319 (2007).
 - [13] F. G. S. L. Brandão, M. Broughton, E. Farhi, S. Gutmann, and H. Napp, arXiv preprint arXiv:1812.04170 10.48550/arxiv.1812.04170 (2018).
 - [14] V. Akshay, D. Rabinovich, E. Campos, and J. Biamonte, *Physical Review A* **104**, L010401 (2021).
 - [15] A. Galda, X. Liu, D. Lykov, Y. Alexeev, and I. Safro, in *2021 IEEE International Conference on Quantum Computing and Engineering (QCE)* (IEEE, 2021) pp. 171–180.
 - [16] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, *Nature Communications* **9**, 4812 (2018).
 - [17] A. Arrasmith, M. Cerezo, P. Czarnik, L. Cincio, and P. J. Coles, *Quantum* **5**, 558 (2021).
 - [18] S. Wang, E. Fontana, M. Cerezo, K. Sharma, A. Sone, L. Cincio, and P. J. Coles, *Nature Communications* **12**, 6961 (2021).
 - [19] D. Kreuzer, D. Beaini, W. L. Hamilton, V. Létourneau, and P. Tossou, in *Advances in Neural Information Processing Systems*, Vol. 34 (2021) pp. 21618–21629.
 - [20] M. Fiedler, *Czechoslovak Mathematical Journal* **23**, 298 (1973).
 - [21] B. Weisfeiler and A. Leman, *NTI, Series 2* **9**, 12 (1968).
 - [22] W. L. Hamilton, R. Ying, and J. Leskovec, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
 - [23] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals, and G. E. Dahl, in *International Conference on Machine Learning* (PMLR, 2017) pp. 1263–1272.
 - [24] D. J. Egger, J. Mareček, and S. Woerner, *Quantum* **5**, 479 (2021).
 - [25] S. H. Sack and M. Serbyn, *Quantum* **5**, 491 (2021).
 - [26] N. Jain, B. Coyle, E. Kashefi, and N. Kumar, *Quantum* **6**, 861 (2022).
 - [27] S. Khairy, R. Shaydulin, L. Cincio, Y. Alexeev, and P. Balaprakash, in *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 34 (2020) pp. 2367–2375.
 - [28] S. Tibaldi, D. Vodola, E. Tignone, and E. Ercolelli, *IEEE Transactions on Quantum Engineering* **4**, 1 (2023).
 - [29] R. Shaydulin and S. M. Wild, arXiv preprint arXiv:2306.16979 10.48550/arxiv.2306.16979 (2023).
 - [30] M. Streif and M. Leib, *Quantum Science and Technology* **5**, 034008 (2020).
 - [31] C. Moussa, H. Calandra, and V. Dunjko, *EPJ Quantum Technology* **9**, 11 (2022).
 - [32] F. Sauvage, S. Sim, A. A. Kunitsa, W. A. Simon, M. Mauri, and A. Perdomo-Ortiz, *Quantum Science and Technology* **9**, 015029 (2024).
 - [33] Y. Bengio, A. Lodi, and A. Prouvost, *European Journal of Operational Research* **290**, 405 (2021).
 - [34] Q. Cappart, D. Chételat, E. B. Khalil, A. Lodi, C. Morris, and P. Veličković, *Journal of Machine Learning Research* **24**, 1 (2023).
 - [35] E. B. Khalil, H. Dai, Y. Zhang, B. Dilkina, and L. Song, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
 - [36] H. Dai, B. Dai, and L. Song, in *International Conference on Machine Learning* (PMLR, 2016) pp. 723–732.
 - [37] W. Kool, H. van Hoof, and M. Welling, in *International Conference on Learning Representations* (2019).
 - [38] M. J. A. Schuetz, J. K. Brubaker, and H. G. Katzgraber, *Nature Machine Intelligence* **4**, 367 (2022).
 - [39] D. A. Nix and A. S. Weigend, in *Proceedings of 1994 IEEE International Conference on Neural Networks*, Vol. 1 (IEEE, 1994) pp. 55–60.
 - [40] A. Kendall and Y. Gal, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
 - [41] B. Lakshminarayanan, A. Pritzel, and C. Blundell, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
 - [42] Y. Gal and Z. Ghahramani, in *International Conference on Machine Learning* (PMLR, 2016) pp. 1050–1059.
 - [43] V. Kuleshov, N. Fenner, and S. Ermon, in *International Conference on Machine Learning* (PMLR, 2018) pp. 2796–2804.
 - [44] M. Abdar, F. Pourpanah, S. Hussain, D. Rezazadegan, L. Liu, M. Ghavamzadeh, P. Fieguth, X. Cao, A. Khos-

- ravi, U. R. Acharya, V. Makarenkov, and S. Nahavandi, *Information Fusion* **76**, 243 (2021).
- [45] E. Hüllermeier and W. Waegeman, *Machine Learning* **110**, 457 (2021).
- [46] A. F. Psaros, X. Meng, Z. Zou, L. Guo, and G. E. Karniadakis, *Journal of Computational Physics* **477**, 111902 (2023).
- [47] D. C. Dowson and B. V. Landau, *Journal of Multivariate Analysis* **12**, 450 (1982).
- [48] M. Arjovsky, S. Chintala, and L. Bottou, in *International Conference on Machine Learning* (PMLR, 2017) pp. 214–223.
- [49] C. Frogner, C. Zhang, H. Mobahi, M. Araya, and T. Poggio, in *Advances in Neural Information Processing Systems*, Vol. 28 (2015).
- [50] C. Villani, *Optimal Transport: Old and New*, Vol. 338 (Springer, 2009).
- [51] M. Cuturi, in *Advances in Neural Information Processing Systems*, Vol. 26 (2013).
- [52] G. Peyré and M. Cuturi, *Foundations and Trends in Machine Learning* **11**, 355 (2019).
- [53] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, in *International Conference on Machine Learning* (PMLR, 2020) pp. 1597–1607.
- [54] A. van den Oord, Y. Li, and O. Vinyals, arXiv preprint arXiv:1807.03748 10.48550/arxiv.1807.03748 (2018).
- [55] Y. You, T. Chen, Y. Sui, T. Chen, Z. Wang, and Y. Shen, in *Advances in Neural Information Processing Systems*, Vol. 33 (2020) pp. 5812–5823.
- [56] S. Suresh, P. Li, C. Hao, and G. Nishi, in *Advances in Neural Information Processing Systems*, Vol. 34 (2021) pp. 15920–15933.
- [57] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, in *Advances in Neural Information Processing Systems*, Vol. 33 (2020) pp. 18661–18673.
- [58] B. Amos, *Foundations and Trends in Machine Learning* **16**, 592 (2023).
- [59] T. Chen, X. Chen, W. Chen, H. Heaton, J. Liu, Z. Wang, and W. Yin, *Journal of Machine Learning Research* **23**, 1 (2022).
- [60] C. Finn, P. Abbeel, and S. Levine, in *International Conference on Machine Learning* (PMLR, 2017) pp. 1126–1135.
- [61] R. Shu, H. H. Xu, T.-H. Oh, and K. Swersky, in *International Conference on Learning Representations* (2022).
- [62] J. L. Ba, J. R. Kiros, and G. E. Hinton, arXiv preprint arXiv:1607.06450 10.48550/arxiv.1607.06450 (2016).
- [63] J. A. Nelder and R. Mead, *The Computer Journal* **7**, 308 (1965).
- [64] G. Verdon, M. Broughton, J. R. McClean, K. J. Sung, R. Babbush, Z. Jiang, H. Neven, and D. Maslov, arXiv preprint arXiv:1907.05415 (2019).
- [65] P. L. Bartlett, D. J. Foster, and M. J. Telgarsky, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017).
- [66] B. Neyshabur, S. Bhojanapalli, and N. Srebro, in *International Conference on Learning Representations* (2018).
- [67] N. Golowich, A. Rakhlin, and O. Shamir, in *Proceedings of the 31st Conference on Learning Theory* (2018) pp. 297–299.
- [68] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World* (Springer, 2005).
- [69] Y. Romano, E. Patterson, and E. Candès, in *Advances in Neural Information Processing Systems*, Vol. 32 (2019).
- [70] A. N. Angelopoulos and S. Bates, arXiv preprint arXiv:2107.07511 10.48550/arxiv.2107.07511 (2021).
- [71] P. Erdős and A. Rényi, *Publicationes Mathematicae Debrecen* **6**, 290 (1959).
- [72] A.-L. Barabási and R. Albert, *Science* **286**, 509 (1999).
- [73] D. J. Watts and S. H. Strogatz, *Nature* **393**, 440 (1998).
- [74] A. R. Conn, N. I. M. Gould, and P. L. Toint, *Trust Region Methods* (SIAM, 2000).
- [75] D. P. Kingma and J. Ba, in *International Conference on Learning Representations* (2015).